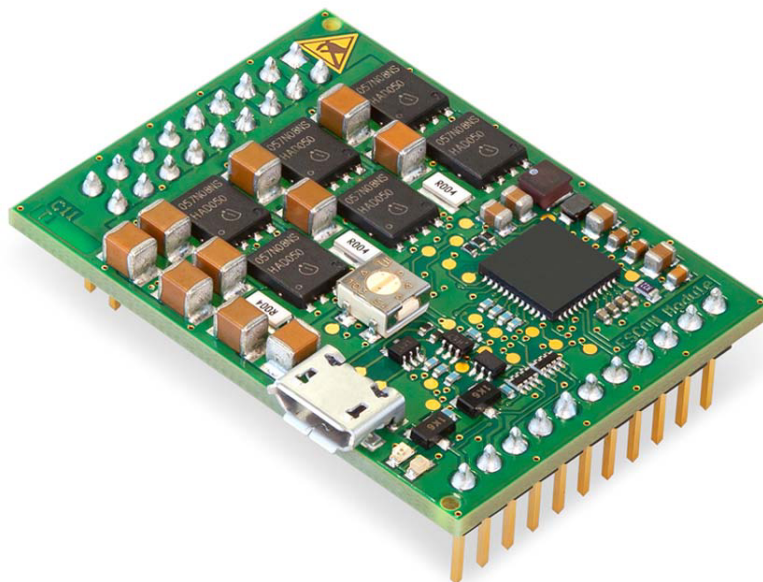


# ***ESCON Module 50/5***

***Servo Controller***

***P/N 438725***

***Hardware Reference***



[escon.maxonmotor.com](http://escon.maxonmotor.com)

***Document ID: rel5744***

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## READ THIS FIRST

**These instructions are intended for qualified technical personnel. Prior commencing with any activities ...**

- you must carefully read and understand this manual and
- you must follow the instructions given therein.

**The ESCON Module 50/5 is considered as partly completed machinery according to EU Directive 2006/42/EC, Article 2, Clause (g) and is intended to be incorporated into or assembled with other machinery or other partly completed machinery or equipment.**

**Therefore, you must not put the device into service, ...**

- unless you have made completely sure that the other machinery fully complies with the EU directive's requirements!
- unless the other machinery fulfills all relevant health and safety aspects!
- unless all respective interfaces have been established and fulfill the herein stated requirements!

# 1 About

## 1.1 About this Document

### 1.1.1 Intended Purpose

The purpose of the present document is to familiarize you with the ESCON Module 50/5 Servo Controller. It will highlight the tasks for safe and adequate installation and/or commissioning. Follow the described instructions ...

- to avoid dangerous situations,
- to keep installation and/or commissioning time at a minimum,
- to increase reliability and service life of the described equipment.

The document contains performance data and specifications, information on fulfilled standards, details on connections and pin assignment, and wiring examples. In addition, the document also includes a Motherboard Design Guide and detailed information on the optionally available «ESCON Module Motherboard».

### 1.1.2 Target Audience

The present document is intended for trained and skilled personnel. It conveys information on how to understand and fulfill the respective work and duties.

### 1.1.3 How to use

Take note of the following notations and codes which will be used throughout the document.

| Notation | Meaning                                                   |
|----------|-----------------------------------------------------------|
| (n)      | refers to an item (such as order number, list item, etc.) |
| →        | denotes “see”, “see also”, “take note of” or “go to”      |

Table 1-1 Notation used

### 1.1.4 Symbols & signs

In the course of the present document, the following symbols and signs will be used.

| Type              | Symbol                                                                                           | Meaning                                                   |                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Safety Alert      | <br>(typical) | DANGER                                                    | Indicates an <b>imminent hazardous situation</b> . If not avoided, it <b>will result in death or serious injury</b> .                        |
|                   |                                                                                                  | WARNING                                                   | Indicates a <b>potential hazardous situation</b> . If not avoided, it <b>may result in death or serious injury</b> .                         |
|                   |                                                                                                  | CAUTION                                                   | Indicates a <b>probably hazardous situation</b> or calls the attention to unsafe practices. If not avoided, it <b>may result in injury</b> . |
| Prohibited Action | <br>(typical) | Indicates a dangerous action. Hence, <b>you must not!</b> |                                                                                                                                              |

| Type                    | Symbol                                                                                         | Meaning                                               |                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Mandatory Action</b> | <br>(typical) | Indicates a mandatory action. Hence, <b>you must!</b> |                                                                                                                            |
| <b>Information</b>      |               | Requirement / Note / Remark                           | Indicates an activity you must perform prior to continuing, or gives information on a particular item you need to observe. |
|                         |               | Best Practice                                         | Indicates advice or a recommendation on the easiest and best way to proceed.                                               |
|                         |               | Material Damage                                       | Indicates instructions on how to prevent damage to the equipment.                                                          |

Table 1-2 Symbols & Signs

## 1.1.5 Trademarks and Brand Names

For easier legibility, registered brand names are listed below and will not be further tagged with their respective trademark. It must be understood that the brands (the list below is not necessarily concluding) are protected by copyright and/or other intellectual property rights even if their legal trademarks are omitted in the later course of this document.

| Brand Name                | Trademark Owner                          |
|---------------------------|------------------------------------------|
| Littelfuse®<br>SMD NANO2® | © Littelfuse, USA-Chicago, IL            |
| Windows®                  | © Microsoft Corporation, USA-Redmond, WA |

Table 1-3 Brand Names and Trademark Owners

## 1.1.6 Copyright

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**maxon motor ag**  
Brünigstrasse 220  
P.O.Box 263  
CH-6072 Sachseln

Phone +41 41 666 15 00  
Fax +41 41 666 16 50  
Web [www.maxonmotor.com](http://www.maxonmotor.com)

## 1.2 About the Device

The ESCON Module 50/5 is a small-sized, powerful 4-quadrant PWM servo controller for the highly efficient control of permanent magnet-activated brushed DC motors or brushless EC motors up to approximately 250 Watts.

The featured operating modes – speed control (closed loop), speed control (open loop), and current control – meet the highest requirements. The ESCON Module 50/5 is designed being commanded by an analog set value and features extensive analog and digital I/O functionality.

The miniaturized OEM plug-in module can be seamlessly integrated in complex customer applications. A suitable motherboard is available for the initial commissioning.

The device is designed to be configured via USB interface using the graphical user interface «ESCON Studio» for Windows PCs.

You can download the latest ESCON software version (as well as the latest edition of the documentation) from the internet under → <http://escon.maxonmotor.com>.

## 1.3 About the Safety Precautions

- Make sure that you have read and understood the note “READ THIS FIRST” on page A-2!
- Do not engage in any work unless you possess the stated skills (→ chapter “1.1.2 Target Audience” on page 1-3)!
- Refer to → chapter “1.1.4 Symbols & signs” on page 1-3 for explanations of the symbols used in the following!
- You must observe any regulation applicable in the country and/or at the site of implementation with regard to health and safety/accident prevention and/or environmental protection!



### DANGER

#### High Voltage and/or Electrical Shock

##### Touching live wires causes death or serious injuries!

- Consider any power cable as connected to live power, unless having proven the opposite!
- Make sure that neither end of cable is connected to live power!
- Make sure that power source cannot be engaged while work is in process!
- Obey lock-out/tag-out procedures!
- Make sure to securely lock any power engaging equipment against unintentional engagement and tag it with your name!



### Requirements

- Make sure that all associated devices and components are installed according to local regulations.
- Be aware that, by principle, an electronic apparatus can not be considered fail-safe. Therefore, you must make sure that any machine/apparatus has been fitted with independent monitoring and safety equipment. If the machine/apparatus should break down, if it is operated incorrectly, if the control unit breaks down or if the cables break or get disconnected, etc., the complete drive system must return – and be kept – in a safe operating mode.
- Be aware that you are not entitled to perform any repair on components supplied by maxon motor.



### Electrostatic Sensitive Device (ESD)

- Make sure to wear working cloth in compliance with ESD.
- Handle device with extra care.

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## 2 Specifications

### 2.1 Technical Data

| ESCON Module 50/5 (438725) |                                                        |                                                                                |
|----------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------|
| Electrical Rating          | Nominal operating voltage $+V_{CC}$                    | 10...50 VDC                                                                    |
|                            | Absolute operating voltage $+V_{CC \min}/+V_{CC \max}$ | 8 VDC/56 VDC                                                                   |
|                            | Output voltage (max.)                                  | $0.98 \times +V_{CC}$                                                          |
|                            | Output current $I_{cont}/I_{max}$ (<20 s)              | 5 A/15 A                                                                       |
|                            | Pulse width modulation frequency                       | 53.6 kHz                                                                       |
|                            | Sampling rate of PI current controller                 | 53.6 kHz                                                                       |
|                            | Sampling rate of PI speed controller                   | 5.36 kHz                                                                       |
|                            | Max. efficiency                                        | 98%                                                                            |
|                            | Max. speed of DC motor                                 | limited by max. permissible speed (motor) and max. output voltage (controller) |
|                            | Max. speed EC motor                                    | 150,000 rpm (1 pole pair)                                                      |
|                            | Built-in motor choke                                   | —                                                                              |
| Inputs & Outputs           | Analog input 1<br>Analog input 2                       | 12-bit resolution; $-10...+10$ V; differential                                 |
|                            | Analog output 1<br>Analog output 2                     | 12-bit resolution ; $-4...+4$ V; referenced to GND                             |
|                            | Digital input 1<br>Digital input 2                     | $+2.4...+36$ VDC ( $R_i = 38.5$ k $\Omega$ )                                   |
|                            | Digital input/output 3<br>Digital input/output 4       | $+2.4...+36$ VDC ( $R_i = 38.5$ k $\Omega$ )/max. 36 VDC ( $I_L < 500$ mA)     |
|                            | Hall sensor signals                                    | H1, H2, H3                                                                     |
|                            | Encoder signals                                        | A, A\, B, B\, (max. 1 MHz)                                                     |
| Voltage Outputs            | Auxiliary output voltage                               | $+5$ VDC ( $I_L \leq 10$ mA)                                                   |
|                            | Hall sensor supply voltage                             | $+5$ VDC ( $I_L \leq 30$ mA)                                                   |
|                            | Encoder supply voltage                                 | $+5$ VDC ( $I_L \leq 70$ mA)                                                   |
| Potentiometer              | Potentiometer P1 (on board)                            | 210°; linear                                                                   |
| Motor Connections          | DC motor                                               | + Motor, – Motor                                                               |
|                            | EC motor                                               | Motor winding 1, Motor winding 2, Motor winding 3                              |
| Interface                  | USB 2.0 / USB 3.0                                      | full speed                                                                     |
| Status Indicators          | Operation                                              | green LED                                                                      |
|                            | Error                                                  | red LED                                                                        |
| Physical                   | Weight                                                 | approx. 12 g                                                                   |
|                            | Dimensions (L x W x H)                                 | 43.2 x 31.8 x 12.7 mm                                                          |
|                            | Connection                                             | Plugs into socket headers with 2.54 mm pitch                                   |

| ESCON Module 50/5 (438725)  |                         |                                      |                                               |
|-----------------------------|-------------------------|--------------------------------------|-----------------------------------------------|
| Environmental<br>Conditions | Temperature             | Operation                            | -30...+45 °C                                  |
|                             |                         | Extended range <sup>*1)</sup>        | +45...+75 °C<br>Derating → Figure 2-1         |
|                             |                         | Storage                              | -40...+85 °C                                  |
|                             | Altitude <sup>*2)</sup> | Operation                            | 0...6'000 m MSL                               |
|                             |                         | Extended range <sup>*1)</sup>        | 6'000...10'000 m MSL<br>Derating → Figure 2-1 |
|                             | Humidity                | 5...90% (condensation not permitted) |                                               |

\*1) Operation within the extended range (temperature and altitude) is permitted. However, a respective derating (declination of output current  $I_{cont}$ ) as to the stated values will apply.

\*2) Operating altitude in meters above Mean Sea Level, MSL.

Table 2-4 Technical Data

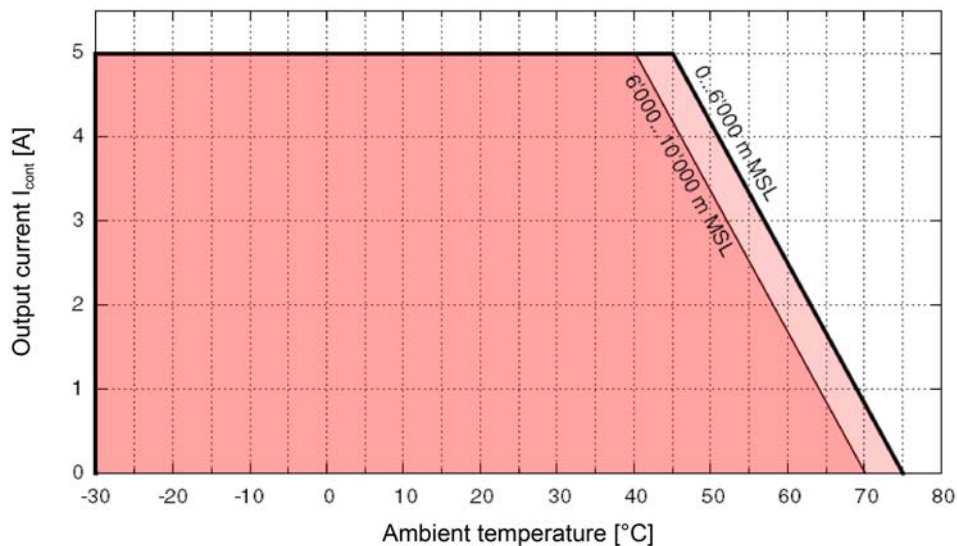


Figure 2-1 Derating Output Current



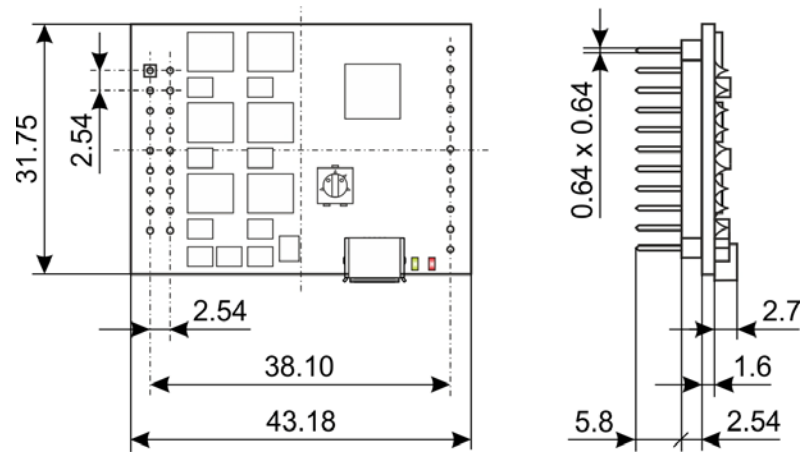


Figure 2-2 Dimensional Drawing [mm]

## 2.2 Standards

The described device has been successfully tested for compliance with the below listed standards. In practical terms, only the complete system (the fully operational equipment comprising all individual components, such as motor, servo controller, power supply unit, EMC filter, cabling etc.) can undergo an EMC test to ensure interference-free operation.



### Important Notice

*The device's compliance with the mentioned standards does not imply its compliance within the final, ready to operate setup. In order to achieve compliance of your operational system, you must perform EMC testing of the involved equipment as a whole.*

| Electromagnetic compatibility |                                                           |                                                                                                                                                                                                                                                        |
|-------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Generic standards             | IEC/EN 61000-6-2                                          | Immunity for industrial environments                                                                                                                                                                                                                   |
|                               | IEC/EN 61000-6-3                                          | Emission standard for residential, commercial and light-industrial environments                                                                                                                                                                        |
| Applied standards             | IEC/EN 61000-6-3<br>IEC/EN 55022<br>(CISPR22)             | Radio disturbance characteristics/radio interference                                                                                                                                                                                                   |
|                               | IEC/EN 61000-4-3                                          | Radiated, radio-frequency, electromagnetic field immunity test >10 V/m                                                                                                                                                                                 |
|                               | IEC/EN 61000-4-4                                          | Electrical fast transient/burst immunity test $\pm 2$ kV                                                                                                                                                                                               |
|                               | IEC/EN 61000-4-6                                          | Immunity to conducted disturbances, induced by radio-frequency fields 10 Vrms                                                                                                                                                                          |
| Others                        |                                                           |                                                                                                                                                                                                                                                        |
| Environmental standards       | IEC/EN 60068-2-6                                          | Environmental testing – Test Fc: Vibration (sinusoidal, 10...500 Hz, 20 m/s <sup>2</sup> )                                                                                                                                                             |
|                               | MIL-STD-810F                                              | Random transport (10...500 Hz up to 2.53 g <sub>rms</sub> )                                                                                                                                                                                            |
| Safety standards              | UL File Number E243951; unassembled printed circuit board |                                                                                                                                                                                                                                                        |
| Reliability                   | MIL-HDBK-217F                                             | Reliability prediction of electronic equipment<br>Environment: Ground, benign (GB)<br>Ambient temperature: 298 K (25 °C)<br>Component stress: In accordance with circuit diagram and nominal power<br>Mean Time Between Failures (MTBF): 639'548 hours |

Table 2-5 Standards

### 3 Setup

**IMPORTANT NOTICE: PREREQUISITES FOR PERMISSION TO COMMENCE INSTALLATION**

The **ESCON Module 50/5** is considered as partly completed machinery according to EU Directive 2006/42/EC, Article 2, Clause (g) and **is intended to be incorporated into or assembled with other machinery or other partly completed machinery or equipment.**

**WARNING****Risk of Injury**

**Operating the device without the full compliance of the surrounding system with EU Directive 2006/42/EC may cause serious injuries!**

- Do not operate the device, unless you have made completely sure that the other machinery fully complies with the EU directive's requirements!
- Do not operate the device, unless the other machinery fulfills all relevant health and safety aspects!
- Do not operate the device, unless all respective interfaces have been established and fulfill the requirements stated in this document!

#### 3.1 Generally applicable Rules

**Maximal permitted Supply Voltage**

- Make sure that operating voltage is between 10...50 VDC.
- Supply voltages above 56 VDC, or wrong polarity will destroy the unit.
- Note that the necessary output current is depending on the load torque. Yet, the output current limits of the ESCON Module 50/5 are as follows; continuous max. 5 A / short-time (acceleration) max. 15 A.

### 3.2 Configuration of Power Supply

Basically, any power supply may be used, provided it meets the minimal requirements stated below.

| Power Supply Requirements |                                                                                                                                                    |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Output voltage            | +V <sub>CC</sub> 10...50 VDC                                                                                                                       |
| Absolute output voltage   | min. 8 VDC; max. 56 VDC                                                                                                                            |
| Output current            | Depending on load <ul style="list-style-type: none"> <li>• continuous max. 5 A</li> <li>• short-time (acceleration, &lt;20 s) max. 15 A</li> </ul> |

- 1) Use the formula below to calculate the required voltage under load.
- 2) Choose a power supply according to the calculated voltage. Thereby consider:
  - a) During braking of the load, the power supply must be capable of buffering the recovered kinetic energy (for example, in a capacitor).
  - b) If you are using an electronically stabilized power supply, make sure that the overcurrent protection circuit is configured inoperative within the operating range.



#### Note

The formula already takes the following into account:

- Maximum PWM duty cycle of 98%
- Controller's max. voltage drop of 1 V @ 5 A

#### KNOWN VALUES:

- Operating torque M [mNm]
- Operating speed n [rpm]
- Nominal motor voltage U<sub>N</sub> [Volt]
- Motor no-load speed at U<sub>N</sub>, n<sub>0</sub> [rpm]
- Speed/torque gradient of the motor Δn/ΔM [rpm/mNm]

#### SOUGHT VALUE:

- Supply voltage +V<sub>CC</sub> [Volt]

#### SOLUTION:

$$V_{CC} \geq \left[ \frac{U_N}{n_0} \cdot \left( n + \frac{\Delta n}{\Delta M} \cdot M \right) \cdot \frac{1}{0.98} \right] + 1 [V]$$

### 3.3 Connections

The actual connection will depend on the overall configuration of your drive system and the type of motor you will be using.

Follow the description in the given order and choose the wiring diagram that best suits the components you are using. For corresponding wiring diagrams → Chapter “4 Wiring” on page 4-25.

#### 3.3.1 Pin Assignment

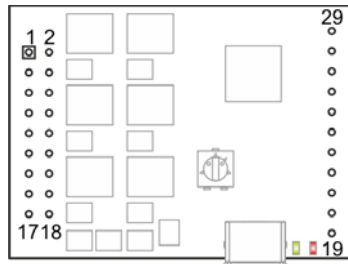


Figure 3-3 Pin Assignment

| Pin  | Signal                        | Description                                                                                                                         |
|------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1/2  | Motor (+M)<br>Motor winding 1 | DC motor: Motor +<br>EC motor: Winding 1                                                                                            |
| 3/4  | Motor (-M)<br>Motor winding 2 | DC motor: Motor -<br>EC motor: Winding 2                                                                                            |
| 5/6  | Motor winding 3               | EC motor: Winding 3                                                                                                                 |
| 7/8  | +V <sub>CC</sub>              | Nominal operating voltage (+10...+50 VDC)                                                                                           |
| 9/10 | Power_GND<br>GND              | Ground of operating voltage<br>Ground                                                                                               |
| 11   | +5 VDC                        | Hall sensor supply voltage (+5 VDC; ≤30 mA)<br>Encoder supply voltage (+5 VDC; ≤70 mA)<br>Auxiliary output voltage (+5 VDC; ≤10 mA) |
| 12   | Channel A                     | Encoder channel A                                                                                                                   |
| 13   | Hall sensor 1                 | Hall sensor 1 input                                                                                                                 |
| 14   | Channel A\                    | Encoder channel A complement                                                                                                        |
| 15   | Hall sensor 2                 | Hall sensor 2 input                                                                                                                 |
| 16   | Channel B                     | Encoder channel B                                                                                                                   |
| 17   | Hall sensor 3                 | Hall sensor 3 input                                                                                                                 |
| 18   | Channel B\                    | Encoder channel B complement                                                                                                        |

Table 3-6 Pin Assignment (Pins 1-18)



Figure 3-4 Pin Assignment

| Pin | Signal        | Description                     |
|-----|---------------|---------------------------------|
| 19  | DigIN/DigOUT4 | Digital input/output 4          |
| 20  | DigIN/DigOUT3 | Digital input/output 3          |
| 21  | DigIN2        | Digital input 2                 |
| 22  | DigIN1        | Digital input 1                 |
| 23  | GND           | Ground                          |
| 24  | AnOUT2        | Analog output 2                 |
| 25  | AnOUT1        | Analog output 1                 |
| 26  | AnIN2–        | Analog input 2, negative signal |
| 27  | AnIN2+        | Analog input 2, positive signal |
| 28  | AnIN1–        | Analog input 1, negative signal |
| 29  | AnIN1+        | Analog input 1, positive signal |

Table 3-7 Pin Assignment (Pins 19-29)

### 3.3.2 Hall Sensor

|                                 |                                       |
|---------------------------------|---------------------------------------|
| Hall sensor supply voltage      | +5 VDC                                |
| Max. Hall sensor supply current | 30 mA                                 |
| Input voltage                   | 0...24 VDC                            |
| Max. input voltage              | +24 VDC                               |
| Logic 0                         | typically <1.0 V                      |
| Logic 1                         | typically >2.4 V                      |
| Internal pull-up resistor       | 10 k $\Omega$ (referenced to +5.45 V) |

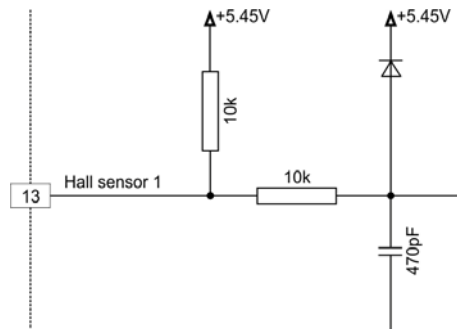


Figure 3-5 Hall Sensor 1 Input Circuit (analogously valid also for Hall Sensors 2 & 3)

### 3.3.3 Encoder



#### Best practice

- Differential signals offer good resistance against electrical interference. Therefore we **recommend using a differential scheme**. Nevertheless, the controller supports both schemes – differential and single-ended (unsymmetrical).
- The controller does not require an index impulse (Ch I, Ch II).
- For best performance, we **strongly recommend using encoders with a line driver**. Otherwise, speed limitations may apply due to slow switching edges.

| Differential                    |                    |
|---------------------------------|--------------------|
| Min. differential input voltage | ±200 mV            |
| Max. input voltage              | +12 VDC/-12 VDC    |
| Line receiver (internal)        | EIA RS422 standard |
| Max. input frequency            | 1 MHz              |

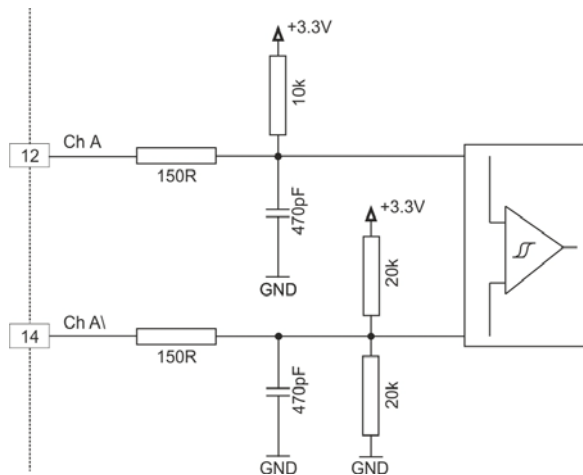


Figure 3-6 Encoder Input Circuit Ch A "Differential" (analogously valid also for Ch B)



| Single-ended         |                                         |
|----------------------|-----------------------------------------|
| Input voltage        | 0...5 VDC                               |
| Max. input voltage   | +12 VDC/-12 VDC                         |
| Logic 0              | <1.0 V                                  |
| Logic 1              | >2.4 V                                  |
| Input high current   | $I_{IH}$ = typically +420 $\mu$ A @ 5 V |
| Input low current    | $I_{IL}$ = typically -170 $\mu$ A @ 0 V |
| Max. input frequency | 100 kHz                                 |

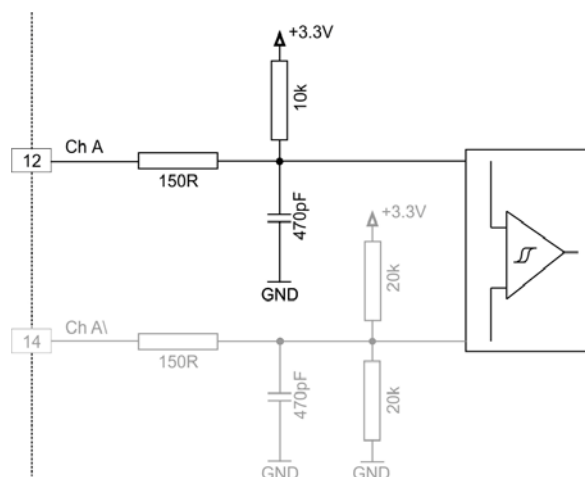


Figure 3-7 Encoder Input Circuit Ch A "Single-ended" (analogously valid also for Ch B)

### 3.3.4 Digital I/Os

#### 3.3.4.1 Digital Input 1

|                          |                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------|
| Input voltage            | 0...36 VDC                                                                                                  |
| Max. input voltage       | +36 VDC/-36 VDC                                                                                             |
| Logic 0                  | typically <1.0 V                                                                                            |
| Logic 1                  | typically >2.4 V                                                                                            |
| Input resistance         | typically 47 k $\Omega$ (<3.3 V)<br>typically 38.5 k $\Omega$ (@ 5 V)<br>typically 25.5 k $\Omega$ (@ 24 V) |
| Input current at logic 1 | typically 130 $\mu$ A @ +5 VDC                                                                              |
| Switching delay          | <8 ms                                                                                                       |

|                                   |                 |
|-----------------------------------|-----------------|
| PWM frequency range               | 10 Hz...5 kHz   |
| PWM duty cycle range (resolution) | 10...90% (0.1%) |
| RC Servo cycle duration           | 3...30 ms       |
| RC Servo pulse length             | 1...2 ms        |

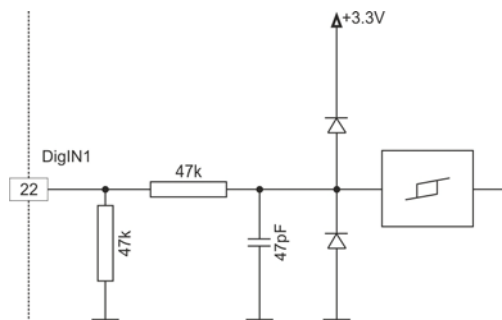


Figure 3-8 DigIN1 Circuit

#### 3.3.4.2 Digital Input 2

|                          |                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------|
| Input voltage            | 0...36 VDC                                                                                                  |
| Max. input voltage       | +36 VDC/-36 VDC                                                                                             |
| Logic 0                  | typically <1.0 V                                                                                            |
| Logic 1                  | typically >2.4 V                                                                                            |
| Input resistance         | typically 47 k $\Omega$ (<3.3 V)<br>typically 38.5 k $\Omega$ (@ 5 V)<br>typically 25.5 k $\Omega$ (@ 24 V) |
| Input current at logic 1 | typically 130 $\mu$ A @ +5 VDC                                                                              |
| Switching delay          | <8 ms                                                                                                       |

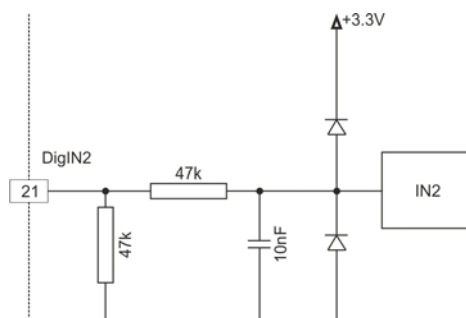


Figure 3-9 DigIN2 Circuit

### 3.3.4.3 Digital Inputs/Outputs 3 and 4

| DigIN                    |                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------|
| Input voltage            | 0...36 VDC                                                                                                  |
| Max. input voltage       | +36 VDC                                                                                                     |
| Logic 0                  | typically <1.0 V                                                                                            |
| Logic 1                  | typically >2.4 V                                                                                            |
| Input resistance         | typically 47 k $\Omega$ (<3.3 V)<br>typically 38.5 k $\Omega$ (@ 5 V)<br>typically 25.5 k $\Omega$ (@ 24 V) |
| Input current at logic 1 | typically 130 $\mu$ A @ +5 VDC                                                                              |
| Switching delay          | <8 ms                                                                                                       |

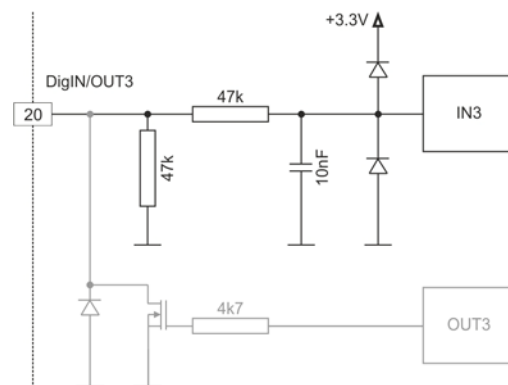


Figure 3-10 DigIN3 Circuit (analogously valid also for DigIN4)

| DigOUT               |                         |
|----------------------|-------------------------|
| Max. input voltage   | +36 VDC                 |
| Max. load current    | 500 mA                  |
| Max. voltage drop    | 0.5 V @ 500 mA          |
| Max. load inductance | 100 mH @ 24 VDC; 500 mA |

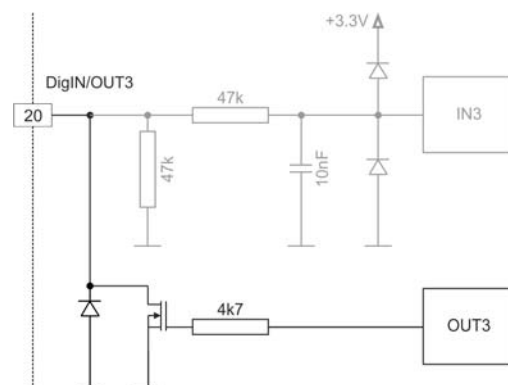


Figure 3-11 DigOUT3 Circuit (analogously valid also for DigOUT4)

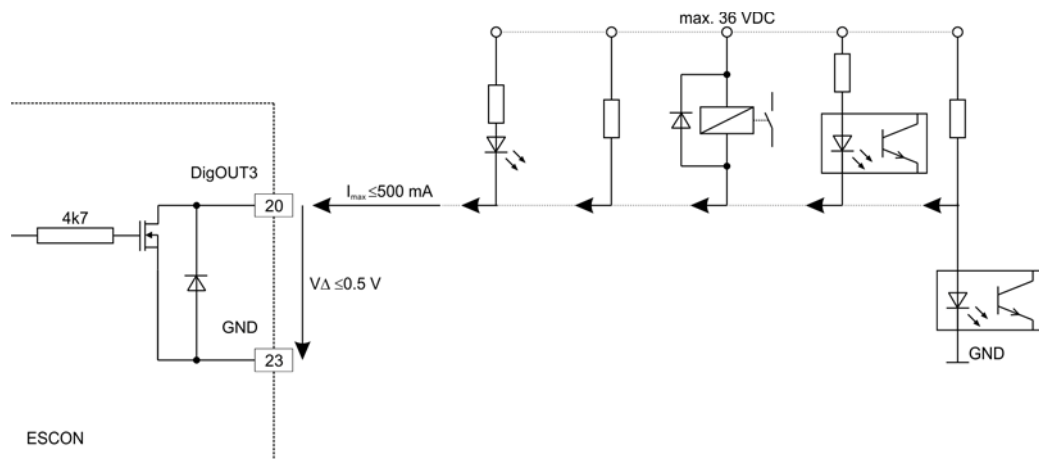


Figure 3-12 DigOUT3 Wiring Examples (analogously valid also for DigOUT4)

### 3.3.5 Analog I/Os

#### 3.3.5.1 Analog Inputs 1 and 2

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| Input voltage       | -10...+10 VDC (differential)                                      |
| Max. input voltage  | +24 VDC/-24 VDC                                                   |
| Common mode voltage | -5...+10 VDC (referenced to GND)                                  |
| Input resistance    | 80 k $\Omega$ (differential)<br>65 k $\Omega$ (referenced to GND) |
| A/D converter       | 12-bit                                                            |
| Resolution          | 5.64 mV                                                           |
| Bandwidth           | 10 kHz                                                            |

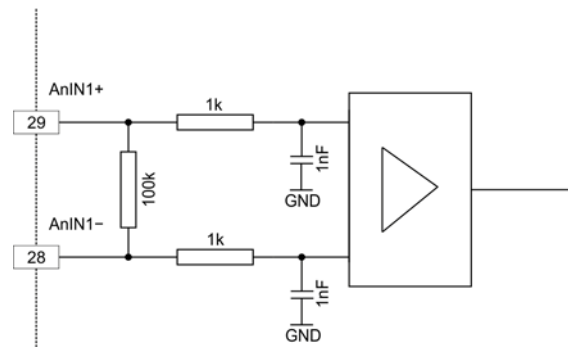


Figure 3-13 AnIN1 Circuit (analogously valid also for AnIN2)

#### 3.3.5.2 Analog Outputs 1 and 2

|                                      |                                                                                                                        |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Output voltage                       | -4...+4 VDC                                                                                                            |
| D/A converter                        | 12-bit                                                                                                                 |
| Resolution                           | 2.42 mV                                                                                                                |
| Refresh rate                         | AnOUT1: 26.8 kHz<br>AnOUT2: 5.4 kHz                                                                                    |
| Analog bandwidth of output amplifier | 50 kHz                                                                                                                 |
| Max. capacitive load                 | 300 nF<br><b>Note:</b> The rate of the increase is limited in proportion to the capacitive load (e.g. 5 V/ms @300 nF). |
| Max. output current limit            | 1 mA                                                                                                                   |

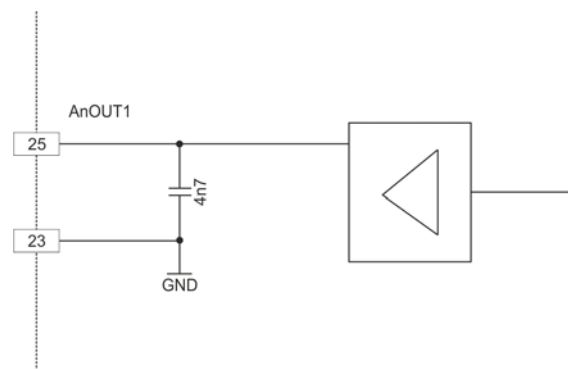


Figure 3-14 AnOUT1 Circuit (analogously valid also for AnOUT2)

### 3.3.6 USB (J7)



Figure 3-15 USB Socket J7


**Note**

Column "Head B" (→Table 3-8) refers to USB terminals of your PC.

| J7 & Head A<br>Pin | Head B<br>Pin | Signal           | Description                         |
|--------------------|---------------|------------------|-------------------------------------|
| 1                  | 1             | V <sub>BUS</sub> | USB BUS supply voltage input +5 VDC |
| 2                  | 2             | D-               | USB Data- (twisted pair with Data+) |
| 3                  | 3             | D+               | USB Data+ (twisted pair with Data-) |
| 4                  | –             | ID               | not connected                       |
| 5                  | 4             | GND              | USB ground                          |

Table 3-8 USB Socket J7 – Pin Assignment & Cabling

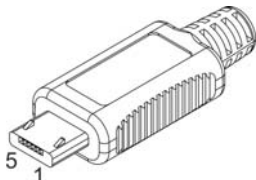
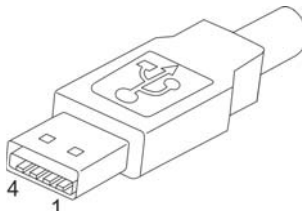
| USB Type A - micro B Cable (403968)                                                                                                                                                                                                |                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div style="display: flex; align-items: center;"> <div style="border: 1px solid black; padding: 2px; margin-right: 5px;"><b>A</b></div>  </div> | <div style="display: flex; align-items: center;">  <div style="border: 1px solid black; padding: 2px; margin-left: 5px;"><b>B</b></div> </div> |
| Cable cross-section                                                                                                                                                                                                                | According to USB 2.0 / USB 3.0 specification                                                                                                                                                                                        |
| Length                                                                                                                                                                                                                             | 1.5 m                                                                                                                                                                                                                               |
| Head A                                                                                                                                                                                                                             | USB type "micro B", male                                                                                                                                                                                                            |
| Head B                                                                                                                                                                                                                             | USB type "A", male                                                                                                                                                                                                                  |

Table 3-9 USB Type A - micro B Cable

|                            |                                |
|----------------------------|--------------------------------|
| USB standard               | USB 2.0 / USB 3.0 (full speed) |
| Max. bus operating voltage | +5.25 VDC                      |
| Typical input current      | 60 mA                          |
| Max. DC data input voltage | –0.5...+3.8 VDC                |

### 3.4 Potentiometer

#### POTENTIOMETER P1

|                  |        |
|------------------|--------|
| Adjustment angle | 210°   |
| Type             | Linear |

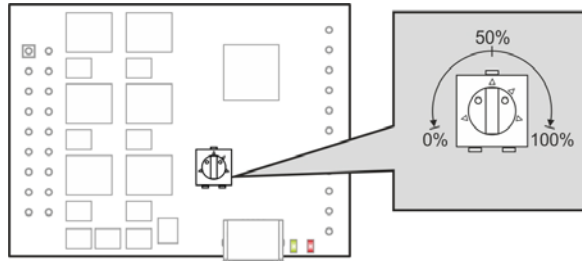


Figure 3-16 Potentiometer – Location & Adjustment Range

### 3.5 Status Indicators

Light-emitting diodes (LEDs) indicate the actual operating status (green) and possible errors (red).

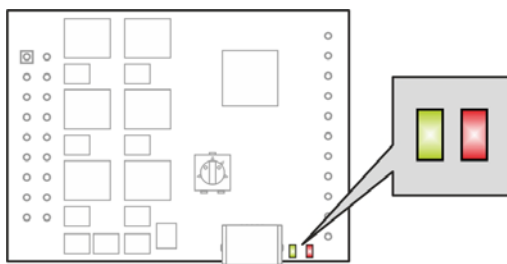


Figure 3-17 LEDs – Location

| LED   |     | Status/Error              |                                                                                                                                                                                          |
|-------|-----|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Green | Red |                           |                                                                                                                                                                                          |
| off   | off | INIT                      |                                                                                                                                                                                          |
| slow  | off | DISABLE                   |                                                                                                                                                                                          |
| on    | off | ENABLE                    |                                                                                                                                                                                          |
| 2x    | off | STOPPING; STOP STANDSTILL |                                                                                                                                                                                          |
| off   | 1x  | ERROR                     | <ul style="list-style-type: none"> <li>• +Vcc Overvoltage Error</li> <li>• +Vcc Undervoltage Error</li> <li>• +5 VDC Undervoltage Error</li> </ul>                                       |
| off   | 2x  | ERROR                     | <ul style="list-style-type: none"> <li>• Thermal Overload Error</li> <li>• Overcurrent Error</li> <li>• Power Stage Protection Error</li> <li>• Internal Hardware Error</li> </ul>       |
| off   | 3x  | ERROR                     | <ul style="list-style-type: none"> <li>• Encoder Cable Break Error</li> <li>• Encoder Polarity Error</li> <li>• DC Tacho Cable Break Error</li> <li>• DC Tacho Polarity Error</li> </ul> |
| off   | 4x  | ERROR                     | • PWM Set Value Input out of Range Error                                                                                                                                                 |
| off   | 5x  | ERROR                     | <ul style="list-style-type: none"> <li>• Hall Sensor Pattern Error</li> <li>• Hall Sensor Sequence Error</li> <li>• Hall Sensor Frequency too high Error</li> </ul>                      |
| off   | on  | ERROR                     | <ul style="list-style-type: none"> <li>• Auto Tuning Identification Error</li> <li>• Internal Software Error</li> </ul>                                                                  |

Timing diagram showing LED states for various error conditions. The diagram has six rows labeled 'slow', '1x', '2x', '3x', '4x', and '5x'. The 'slow' row shows a green LED pulse followed by a red LED pulse, with a 1s interval marked. The '1x' through '5x' rows show a green LED pulse followed by a red LED pulse. The 'on' and 'off' labels are at the end of the diagram.

Table 3-10 LEDs – Interpretation of Condition



## 4 Wiring

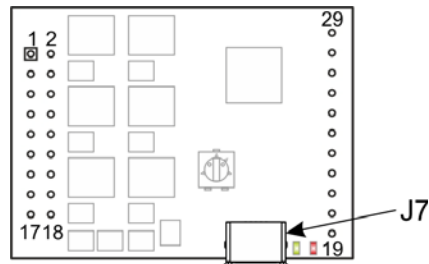


Figure 4-18 Interfaces – Designations and Location



### Note

The subsequent diagrams feature this symbol:

-  Ground safety earth connection (optional)

## 4.1 DC Motors

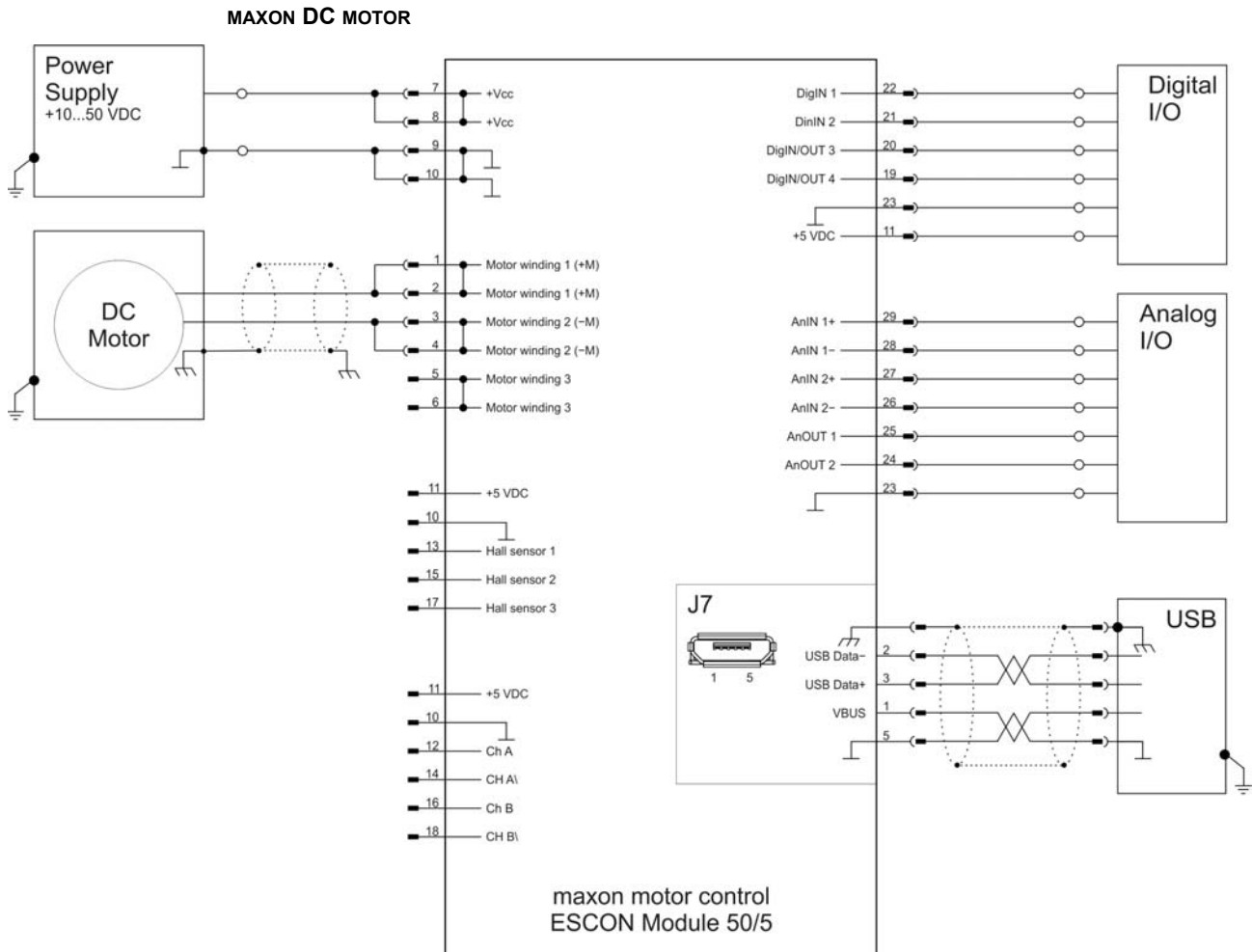


Figure 4-19 maxon DC motor

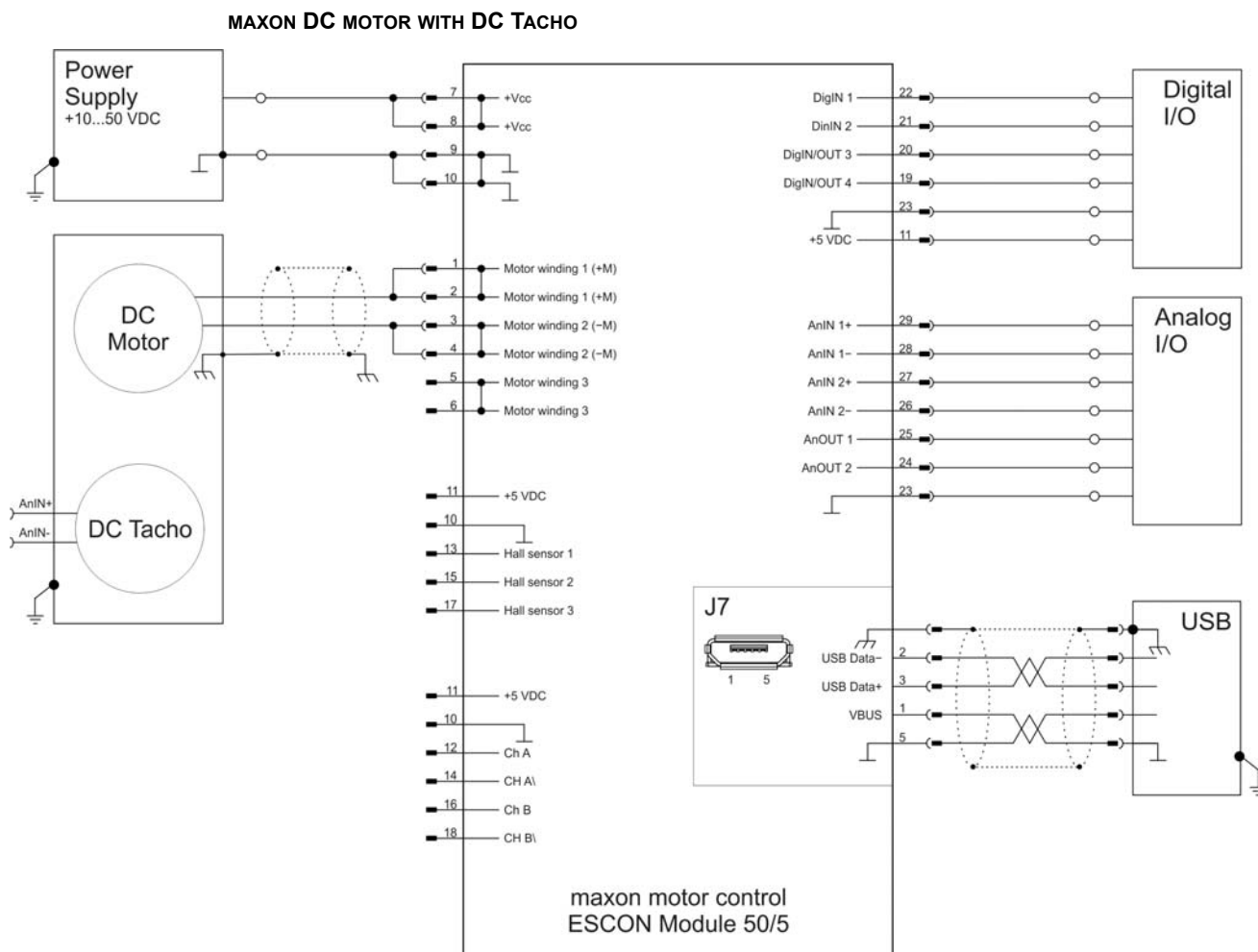


Figure 4-20 maxon DC motor with DC Tacho

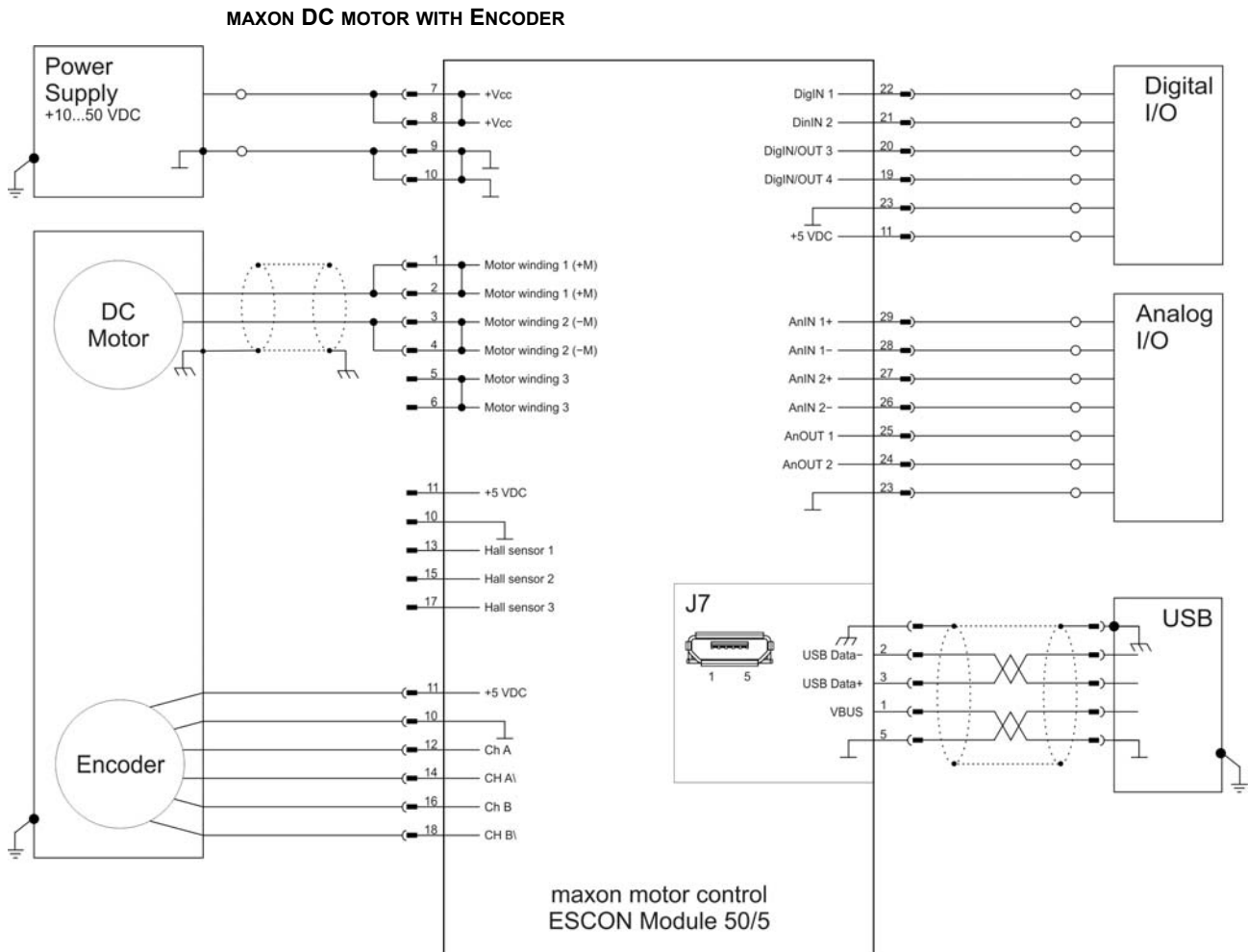


Figure 4-21 maxon DC motor with Encoder

## 4.2 EC Motors

### MAXON EC MOTOR WITH HALL SENSORS

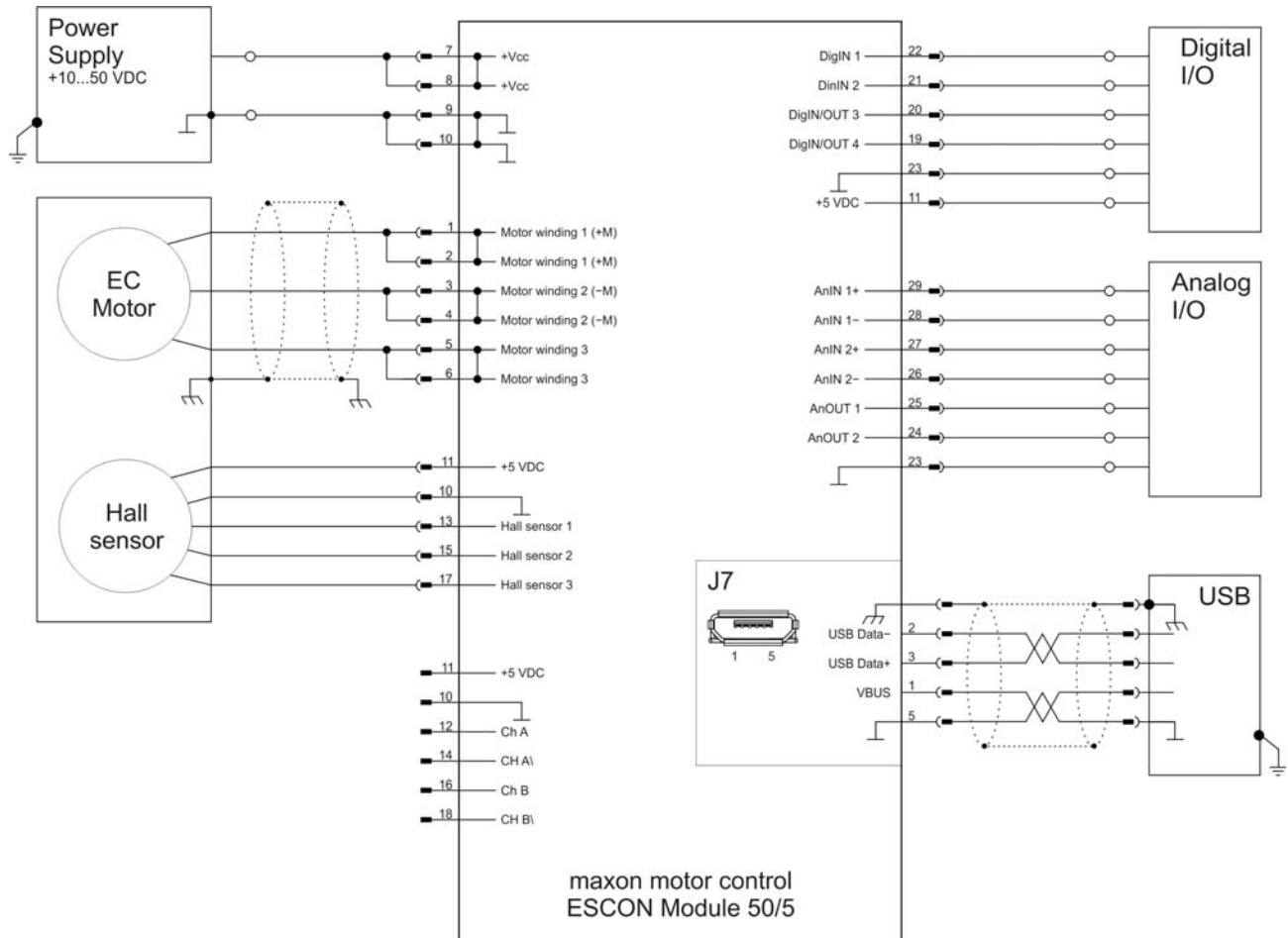


Figure 4-22 maxon EC motor with Hall Sensors

## MAXON EC MOTOR WITH HALL SENSORS & ENCODER

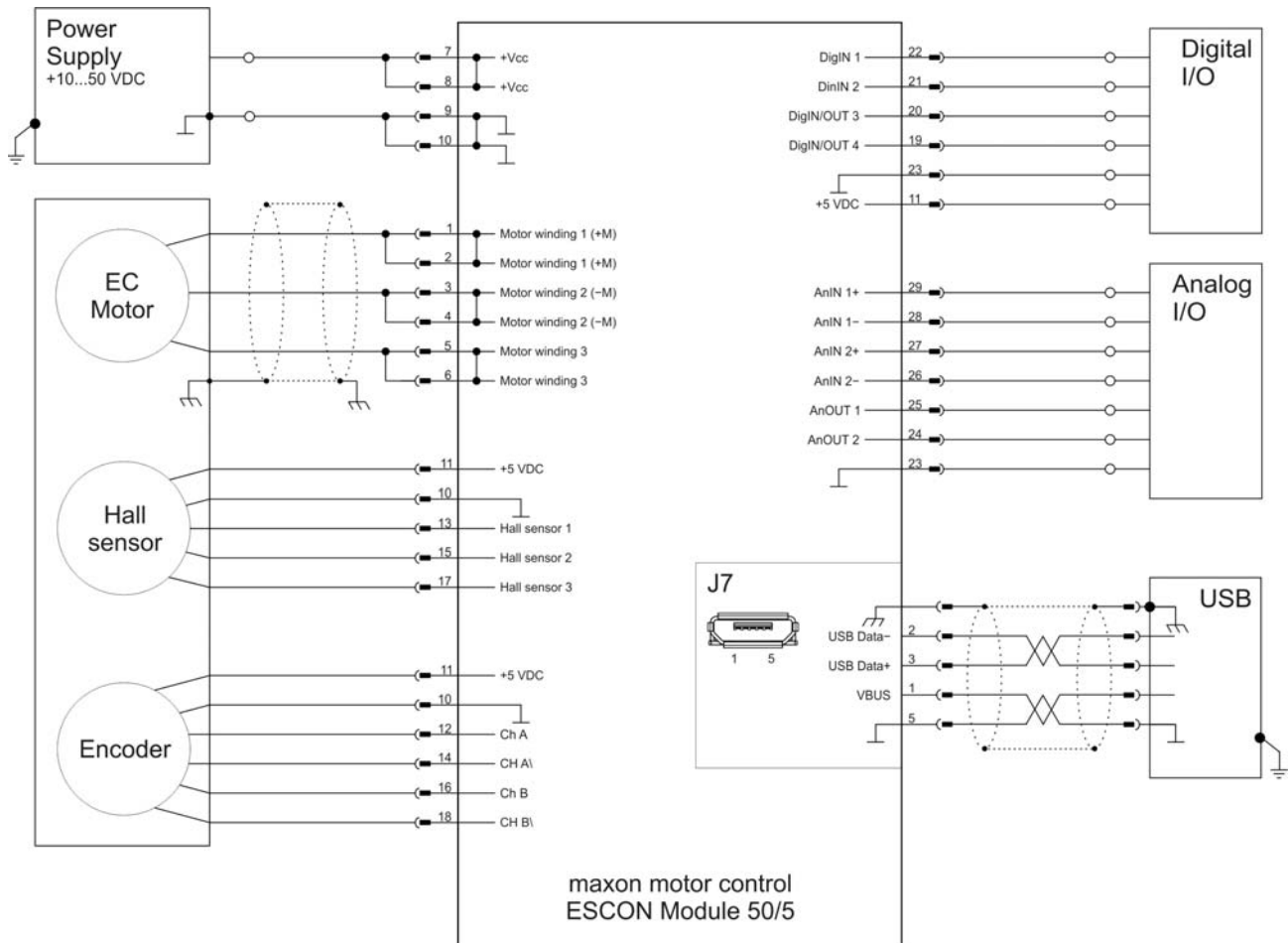


Figure 4-23 maxon EC motor with Hall Sensors & Encoder

## 5 Motherboard Design Guide

The following provides helpful information on integrating the ESCON Module 50/5 on a printed circuit board. The «Motherboard Design Guide» contains recommendations for the layout of the motherboard and specifies external components that may be required, pin assignments, and connection examples.



### CAUTION

#### ***Dangerous Action***

#### ***Errors in implementing the Design can result in serious Injury!***

- *Only proceed if you are skilled in electronics design!*
- *Designing a printed circuit board requires special skills and knowledge and may only be performed by experienced electronic developers!*
- *This quick guide is only intended as an aid, does not make any claim to completeness, and will not automatically result in a functional component!*



#### ***Bring in additional Support:***

*If you are not trained in the design and development of printed circuit boards, you will need additional support for this point.*

*maxon motor will be happy to provide you with a quote for designing and manufacturing a motherboard for your specific application.*

## 5.1 Requirements for Components of Third-party Suppliers

### 5.1.1 Socket Headers

The ESCON Module 50/5's implementation with pin headers permits mounting in two different ways. The module can either be plugged onto a socket header (→ Table 5-11) or be directly soldered to a printed circuit board.

### 5.1.2 Supply Voltage

To protect the ESCON Module 50/5, we recommend using an external circuit breaker, a TVS diode, and a capacitor in the voltage supply cable. In this regard, please note the following recommendations:

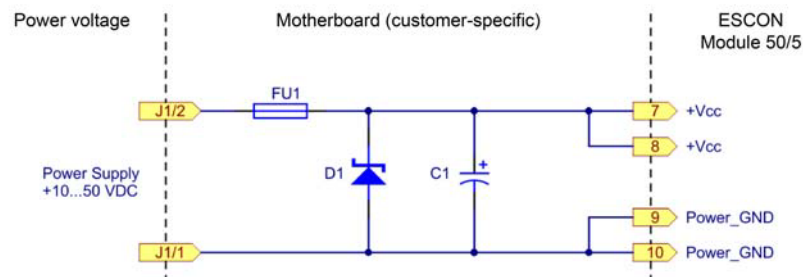


Figure 5-24 Wiring of Voltage Supply Cable

#### INPUT FUSE (FU1)

An input fuse (FU1) is necessary in order to provide reverse polarity protection. Together with an unipolar TVS diode (D1), this prevents current from flowing in the wrong direction.

#### TVS DIODE (D1)

To protect against overvoltage resulting from voltage transients or brake energy feedback, we recommend connecting a TVS (transient voltage suppressor) diode (D1) to the voltage supply cable.

#### CAPACITOR (C1)

The function of the ESCON Module 50/5 does not necessarily require the use of an external capacitor (C1). To further reduce voltage ripple and feedback currents, an electrolytic capacitor can be connected to the voltage supply cable.



### 5.1.3 Motor Cables/Motor Chokes

The ESCON Module 50/5 is not equipped with internal motor chokes.

The majority of motors and applications do not require additional chokes. However, in the case of high supply voltage with very low terminal inductance, the ripple of the motor current can reach an unacceptably high value. This causes the motor to heat up unnecessarily and causes instable control behavior. The minimum terminal inductance required per phase can be calculated using the following formula:

$$L_{phase} \geq \frac{1}{2} \cdot \left( \frac{V_{cc}}{6 \cdot f_{PWM} \cdot I_N} - (0.3 \cdot L_{motor}) \right)$$

|                |                                                                  |
|----------------|------------------------------------------------------------------|
| $L_{phase}[H]$ | Additional external inductance per phase                         |
| $V_{cc}[V]$    | Operating voltage +V <sub>cc</sub>                               |
| $f_{PWM}[Hz]$  | Switching frequency of the power stage = 53 600 Hz               |
| $I_N[A]$       | Nominal current of the motor (→line 6 in the maxon catalog)      |
| $L_{motor}[H]$ | Terminal inductance of the motor (→line 11 in the maxon catalog) |

If the result of the calculation is negative, no additional chokes are necessary. Nevertheless, the use of chokes in combination with additional filter components can be useful to reduce the emission of electromagnetic interference.

An additional choke must feature electromagnetic shielding, a high saturation current, minimal losses, and a nominal current greater than the continuous current of the motor. The below wiring example refers to an additional inductance of 22 µH. If a different additional inductance is required, also the filter components must be adapted accordingly. Should you need further help with the filter design, contact maxon Support at →<http://support.maxonmotor.com>.

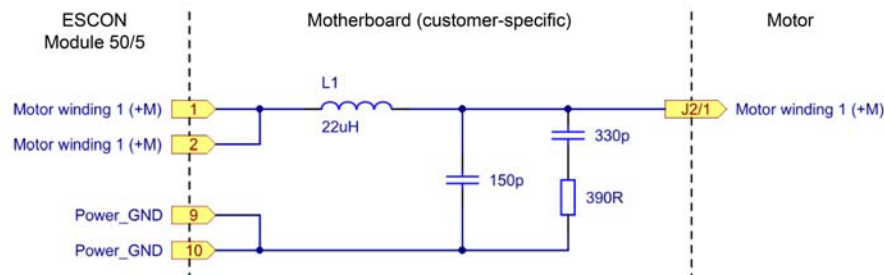


Figure 5-25 Wiring of Motor Winding 1 (analogously valid also for Motor Windings 2 & 3)

## 5.1.4 Recommended Components and Manufacturers

| Recommended components     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Socket Header              | Straight socket header, pluggable with 0.64 x 0.64 mm pin headers, 2.54 mm pitch, 3 A, contact material: gold                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                       |
|                            | 9 poles, 2 rows                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Preci-Dip (803-87-018-10-005101)<br>Würth (613 018 218 21)<br>E-tec (BL2-018-S842-55) |
|                            | 11 poles, 1 row                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Preci-Dip (801-87-011-10-005101)<br>Würth (613 011 118 21)<br>E-tec (BL1-011-S842-55) |
| FU1 Fuse                   | Littelfuse 157 series, fuse holder incl. SMD NANO2<br>Fuse 10 A very fast-acting, 26.46 A <sup>2</sup> sec (0157010.DR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                       |
| TVS Diode D1               | <ul style="list-style-type: none"> <li>Vishay (SMBJ54A)<br/><math>U_R = 54 \text{ V}</math>, <math>U_{BR} = 60.0 \dots 66.3 \text{ V @ } 1 \text{ mA}</math>, <math>U_C = 87.1 \text{ V @ } 6.9 \text{ A}</math></li> <li>Diodec (P6SMBJ54A)<br/><math>U_R = 54 \text{ V}</math>, <math>U_{BR} = 60.0 \dots 66.6 \text{ V @ } 1 \text{ mA}</math>, <math>U_C = 87.1 \text{ V @ } 6.9 \text{ A}</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                       |
| Capacitor C1               | <ul style="list-style-type: none"> <li>Panasonic (EEUFC1J221S)<br/>Rated voltage 63 V, capacitance 220 <math>\mu\text{F}</math>, ripple current 1285 mA</li> <li>Rubycon (63ZL220M10X23)<br/>Rated voltage 63 V, capacitance 220 <math>\mu\text{F}</math>, ripple current 1120 mA</li> <li>Nichicon (UPM1J221MHD)<br/>Rated voltage 63 V, capacitance 220 <math>\mu\text{F}</math>, ripple current 1300 mA</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                       |
| Motor Cable<br>Motor Choke | <ul style="list-style-type: none"> <li>Würth Elektronik WE-PD-XXL (7447709220)<br/><math>L_N = 22 \mu\text{H}</math>, <math>R_{DC} = 23.3 \text{ m}\Omega</math>, <math>I_{DC} = 5.3 \text{ A}</math>, <math>I_{sat} = 6.5 \text{ A}</math>, shielded</li> <li>Coiltronics (DR127-220)<br/><math>L_N = 22 \mu\text{H}</math>, <math>R_{DC} = 39.1 \text{ m}\Omega</math>, <math>I_{DC} = 4.0 \text{ A}</math>, <math>I_{sat} = 7.6 \text{ A}</math>, shielded</li> <li>Würth Elektronik WE-PD-XXL (7447709150)<br/><math>L_N = 15 \mu\text{H}</math>, <math>R_{DC} = 21 \text{ m}\Omega</math>, <math>I_{DC} = 6.5 \text{ A}</math>, <math>I_{sat} = 8.0 \text{ A}</math>, shielded</li> <li>Sumida (CDRH129RNP-150MC)<br/><math>L_N = 15 \mu\text{H}</math>, <math>R_{DC} = 16 \text{ m}\Omega</math>, <math>I_{DC} = 6.0 \text{ A}</math>, <math>I_{sat} &gt; 6.0 \text{ A}</math>, shielded</li> <li>Coiltronics (DR127-150)<br/><math>L_N = 15 \mu\text{H}</math>, <math>R_{DC} = 25 \text{ m}\Omega</math>, <math>I_{DC} = 5.0 \text{ A}</math>, <math>I_{sat} = 9.7 \text{ A}</math>, shielded</li> <li>Bourns (SRR1280-150M)<br/><math>L_N = 15 \mu\text{H}</math>, <math>R_{DC} = 28 \text{ m}\Omega</math>, <math>I_{DC} = 5.2 \text{ A}</math>, <math>I_{sat} &gt; 5.2 \text{ A}</math>, shielded</li> <li>Würth Elektronik WE-PD-XL (744770115)<br/><math>L_N = 15 \mu\text{H}</math>, <math>R_{DC} = 24 \text{ m}\Omega</math>, <math>I_{DC} = 5.0 \text{ A}</math>, <math>I_{sat} = 6.0 \text{ A}</math>, shielded</li> <li>Sumida (CDR127/LDNP-150M)<br/><math>L_N = 15 \mu\text{H}</math>, <math>R_{DC} = 20 \text{ m}\Omega</math>, <math>I_{DC} = 5.7 \text{ A}</math>, <math>I_{sat} &gt; 5.7 \text{ A}</math>, shielded</li> </ul> |                                                                                       |

Table 5-11 Motherboard Design Guide – Recommended Components

## 5.2 Design Guidelines

The following instructions are intended to serve as an aid for designing an application-specific motherboard and ensuring the correct and reliable integration of the ESCON Module 50/5.

### 5.2.1 Ground

All ground connections (GND) should be internally connected to the ESCON Module 50/5 (equal potential). It is customary to equip the motherboard with a ground plane. All ground connections should be connected to the voltage supply ground via wide conductive tracks.

| Pin | Signal           | Description                           |
|-----|------------------|---------------------------------------|
| 9   | Power_GND<br>GND | Ground of operating voltage<br>Ground |
| 10  | Power_GND<br>GND | Ground of operating voltage<br>Ground |
| 23  | GND              | Ground                                |

Table 5-12 Motherboard Design Guide – Grounding

If an earth potential is in place or required, the ground plane should be connected to the earth potential via one or more capacitors. The use of ceramic capacitors with 100 nF and 100 V is recommended.

### 5.2.2 Layout

Guidelines for the layout of the motherboard:

- Connector pins [7] and [8] +V<sub>CC</sub> operating voltage:  
The pins should be connected to the fuse via wide conductive tracks.
- Connector pins [9], [10] and [23] ground:  
All pins should be connected with the ground of the operating voltage via wide conductive tracks.
- The width of the conductive track and the copper coating thickness of the conductors for supply voltage and motor depend on the current required for the application. A minimum width of 75 mil is recommended for the track and a minimum thickness of 35 µm for the copper coating.

## 5.3 THT Footprint

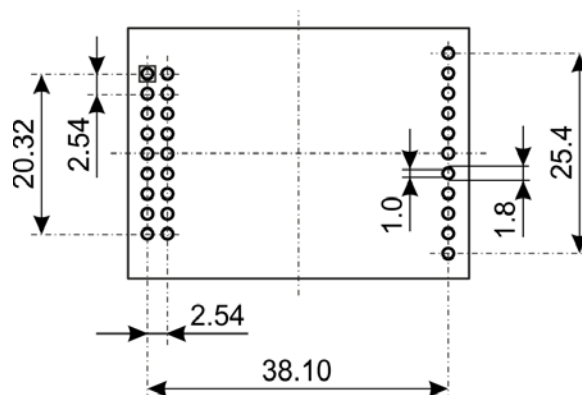


Figure 5-26 THT Footprint [mm] – View from above

## **5.4 Pin Assignment**

For detailed specifications →chapter “3.3 Connections” on page 3-13.

## **5.5 Technical Data**

For detailed specifications →chapter “2 Specifications” on page 2-7.

## **5.6 Dimensional Drawing**

For the dimensional drawing →Figure 2-2 on page 2-9.

## 5.7 ESCON Module Motherboard (438779)

The ESCON Module Motherboard (subsequently named ESCON Module MoBo) is available as an alternative to developing an own motherboard. All required connections are already in place and designed as screw-type terminals.

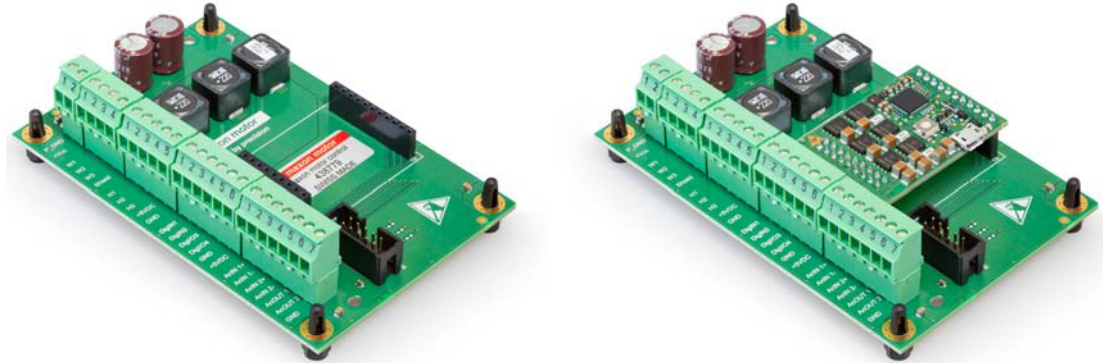


Figure 5-27 ESCON Module MoBo (left), with mounted ESCON Module 50/5 (right)

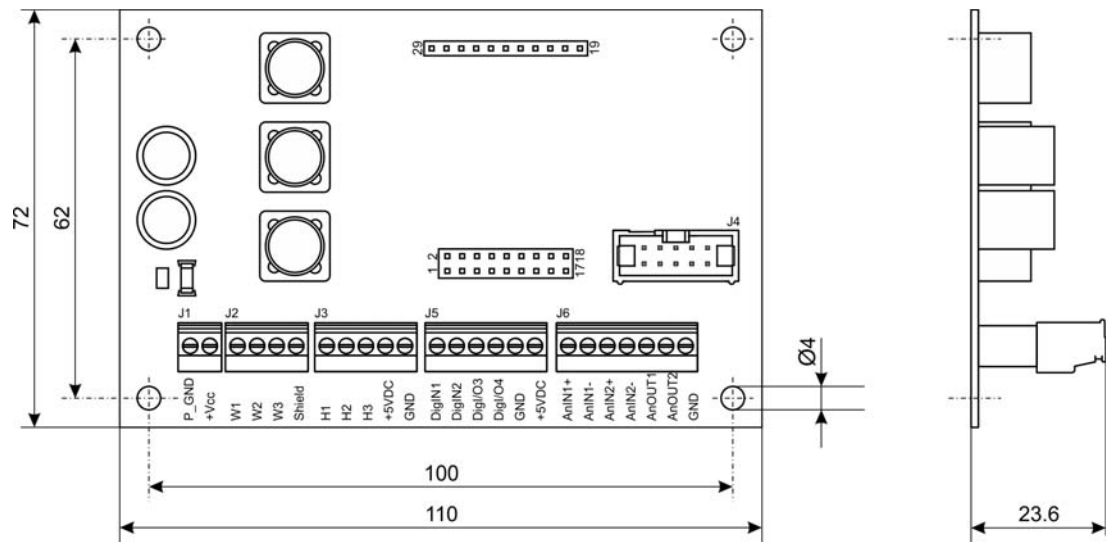


Figure 5-28 ESCON Module MoBo – Dimensional Drawing [mm]

### 5.7.1 Assembly

The ESCON Module MoBo is designed to easily be screw-mounted or integrated into standard rail systems. For ordering information for the components required →Figure 5-29 (only for illustrative purposes) and →Table 5-13.

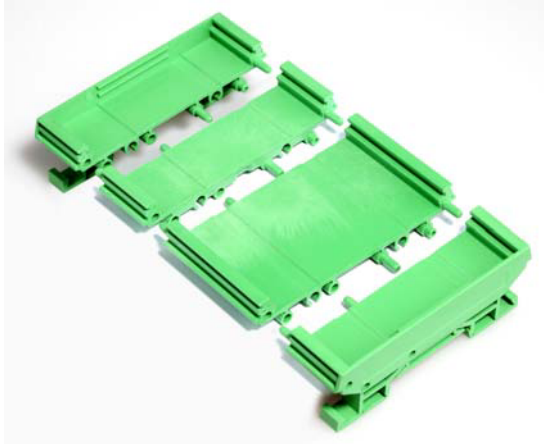


Figure 5-29 ESCON Module MoBo – Mounting on DIN Rail

| Specification / Accessories |                                                                                                                                                                              |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adapter for DIN rail        | <b>PHOENIX CONTACT</b><br>2 x panel mounting base element 11.25 mm UMK-SE11.25-1 (2970442)<br>2 x base element 45 mm UMK-BE45 (2970015)<br>2 x foot element UMK-FE (2970031) |
|                             | <b>CamdenBoss</b><br>2 x end section with foot 22.5 mm (CIME/M/SEF2250S)<br>1 x base element 22.5 mm (CIME/M/BE2250SS)<br>1 x base element 45 mm (CIME/M/BE4500SS)           |

Table 5-13 ESCON Module MoBo, mounting on DIN Rail – Specification & Accessories

## 5.7.2 Connections

**Note**

The USB interface is located directly at the ESCON Module 50/5.

## 5.7.2.1 Power Supply (J1)



Figure 5-30 ESCON Module MoBo – Power Plug J1

| J1<br>Pin | Signal           | Description                               |
|-----------|------------------|-------------------------------------------|
| 1         | Power_GND        | Ground of operating voltage               |
| 2         | +V <sub>CC</sub> | Nominal operating voltage (+10...+50 VDC) |

Table 5-14 ESCON Module MoBo – Power Plug J1 – Pin Assignment

| Specification / Accessories |                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Type                        | Pluggable screw-type terminal block, 2 poles, 3.5 mm pitch                                            |
| Suitable cables             | 0.14...1.5 mm <sup>2</sup> multi-core, AWG 28-14<br>0.14...1.5 mm <sup>2</sup> single wire, AWG 28-14 |

Table 5-15 ESCON Module MoBo – Power Plug J1 – Specification &amp; Accessories

## 5.7.2.2 Motor (J2)

The servo controller is set to drive either maxon DC motors (brushed) or maxon EC motors (brushless).

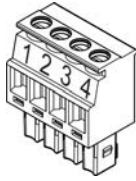


Figure 5-31 ESCON Module MoBo – Motor Plug J2

| J2 Pin | Signal        | Description       |
|--------|---------------|-------------------|
| 1      | Motor (+M)    | DC motor: Motor + |
| 2      | Motor (–M)    | DC motor: Motor – |
| 3      | not connected | –                 |
| 4      | Motor shield  | Cable shield      |

Table 5-16 ESCON Module MoBo – Motor Plug J2 – Pin Assignment for maxon DC motor (brushed)

| J2 Pin | Signal          | Description         |
|--------|-----------------|---------------------|
| 1      | Motor winding 1 | EC motor: Winding 1 |
| 2      | Motor winding 2 | EC motor: Winding 2 |
| 3      | Motor winding 3 | EC motor: Winding 3 |
| 4      | Motor shield    | Cable shield        |

Table 5-17 ESCON Module MoBo – Motor Plug J2 – Pin Assignment for maxon EC motor (brushless)

| Specification / Accessories |                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Type                        | Pluggable screw-type terminal block, 4 poles, 3.5 mm pitch                                            |
| Suitable cables             | 0.14...1.5 mm <sup>2</sup> multi-core, AWG 28-14<br>0.14...1.5 mm <sup>2</sup> single wire, AWG 28-14 |

Table 5-18 ESCON Module MoBo – Motor Plug J2 – Specification & Accessories



### 5.7.2.3 Hall sensor (J3)

Suitable Hall effect sensors IC use «Schmitt trigger» with open collector output.

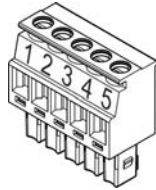


Figure 5-32 ESCON Module MoBo – Hall Sensor Plug J3

| J3<br>Pin | Signal        | Description                                           |
|-----------|---------------|-------------------------------------------------------|
| 1         | Hall sensor 1 | Hall sensor 1 input                                   |
| 2         | Hall sensor 2 | Hall sensor 2 input                                   |
| 3         | Hall sensor 3 | Hall sensor 3 input                                   |
| 4         | +5 VDC        | Hall sensor supply voltage (+5 VDC; $I_L \leq 30$ mA) |
| 5         | GND           | Ground                                                |

Table 5-19 ESCON Module MoBo – Hall Sensor Plug J3 – Pin Assignment

| Specification / Accessories |                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Type                        | Pluggable screw-type terminal block, 5 poles, 3.5 mm pitch                                            |
| Suitable cables             | 0.14...1.5 mm <sup>2</sup> multi-core, AWG 28-14<br>0.14...1.5 mm <sup>2</sup> single wire, AWG 28-14 |

Table 5-20 ESCON Module MoBo – Hall Sensor Plug J3 – Specification & Accessories

#### 5.7.2.4 Encoder (J4)

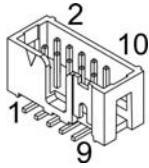


Figure 5-33 ESCON Module MoBo – Encoder Socket J4

| J4 Pin | Signal        | Description                             |
|--------|---------------|-----------------------------------------|
| 1      | not connected | –                                       |
| 2      | +5 VDC        | Encoder supply voltage (+5 VDC; ≤70 mA) |
| 3      | GND           | Ground                                  |
| 4      | not connected | –                                       |
| 5      | Channel A\    | Channel A complement                    |
| 6      | Channel A     | Channel A                               |
| 7      | Channel B\    | Channel B complement                    |
| 8      | Channel B     | Channel B                               |
| 9      | not connected | –                                       |
| 10     | not connected | –                                       |

Table 5-21 ESCON Module MoBo – Encoder Socket J4 – Pin Assignment

| Accessories            |          |                                                                                      |
|------------------------|----------|--------------------------------------------------------------------------------------|
| Suitable strain relief | Retainer | For sockets with strain relief:<br>1 retainer clip, height 13.5 mm, 3M (3505-8110)   |
|                        |          | For sockets without strain relief:<br>1 retainer clip, height 7.9 mm, 3M (3505-8010) |
|                        | Latch    | For sockets with strain relief: 2 pieces, 3M (3505-33B)                              |

Table 5-22 ESCON Module MoBo – Encoder Socket J4 – Accessories

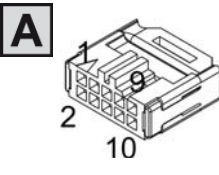
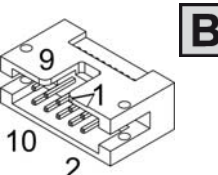
| Encoder Cable (275934)                                                            |                                                                                     |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |
| Cable cross-section                                                               | 10 x AWG28, round-jacket, twisted pair flat cable, 1.27 mm pitch                    |
| Length                                                                            | 3.20 m                                                                              |
| Head A                                                                            | DIN 41651 female, 2.54 mm pitch, 10 pins, with strain relief                        |
| Head B                                                                            | DIN 41651 connector, 2.54 mm pitch, 10 poles, with strain relief                    |

Table 5-23 ESCON Module MoBo – Encoder Cable



#### Best practice

- Because of its resistance against electrical interferences, **we recommend using differential scheme**. Nevertheless, the controller supports both schemes – differential and single-ended.
- The controller does not require an index impulse (Ch I, Ch II).
- For best performance, **we strongly recommend using encoders with line driver**. Otherwise, speed limitations may apply due to slow switching edges.

### 5.7.2.5 Digital I/Os (J5)

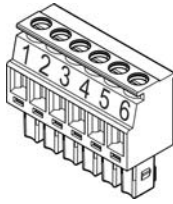


Figure 5-34 ESCON Module MoBo – Digital I/Os Plug J5

| J5<br>Pin | Signal        | Description                               |
|-----------|---------------|-------------------------------------------|
| 1         | DigIN1        | Digital input 1                           |
| 2         | DigIN2        | Digital input 2                           |
| 3         | DigIN/DigOUT3 | Digital input/output 3                    |
| 4         | DigIN/DigOUT4 | Digital input/output 4                    |
| 5         | GND           | Ground                                    |
| 6         | +5 VDC        | Auxiliary output voltage (+5 VDC; ≤10 mA) |

Table 5-24 ESCON Module MoBo – Digital I/Os Plug J5 – Pin Assignment

| Specification / Accessories |                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Type                        | Pluggable screw-type terminal block, 6 poles, 3.5 mm pitch                                            |
| Suitable cables             | 0.14...1.5 mm <sup>2</sup> multi-core, AWG 28-14<br>0.14...1.5 mm <sup>2</sup> single wire, AWG 28-14 |

Table 5-25 ESCON Module MoBo – Digital I/Os Plug J5 – Specification & Accessories

## 5.7.2.6 Analog I/Os (J6)

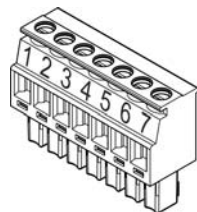


Figure 5-35 ESCON Module MoBo – Analog I/Os Plug J6

| J6<br>Pin | Signal | Description                     |
|-----------|--------|---------------------------------|
| 1         | AnIN1+ | Analog input 1, positive signal |
| 2         | AnIN1– | Analog input 1, negative signal |
| 3         | AnIN2+ | Analog input 2, positive signal |
| 4         | AnIN2– | Analog input 2, negative signal |
| 5         | AnOUT1 | Analog output 1                 |
| 6         | AnOUT2 | Analog output 2                 |
| 7         | GND    | Ground                          |

Table 5-26 ESCON Module MoBo – Analog I/Os Plug J6 – Pin Assignment

| Specification / Accessories |                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Type                        | Pluggable screw-type terminal block, 7 poles, 3.5 mm pitch                                            |
| Suitable cables             | 0.14...1.5 mm <sup>2</sup> multi-core, AWG 28-14<br>0.14...1.5 mm <sup>2</sup> single wire, AWG 28-14 |

Table 5-27 ESCON Module MoBo – Analog I/Os Plug J6 – Specification &amp; Accessories

## 5.7.3 Wiring



### Note

The USB interface is located directly at the ESCON Module 50/5.



### Note

The subsequent diagrams feature this symbol:

- Ground safety earth connection (optional)

### 5.7.3.1 DC Motors

#### MAXON DC MOTOR

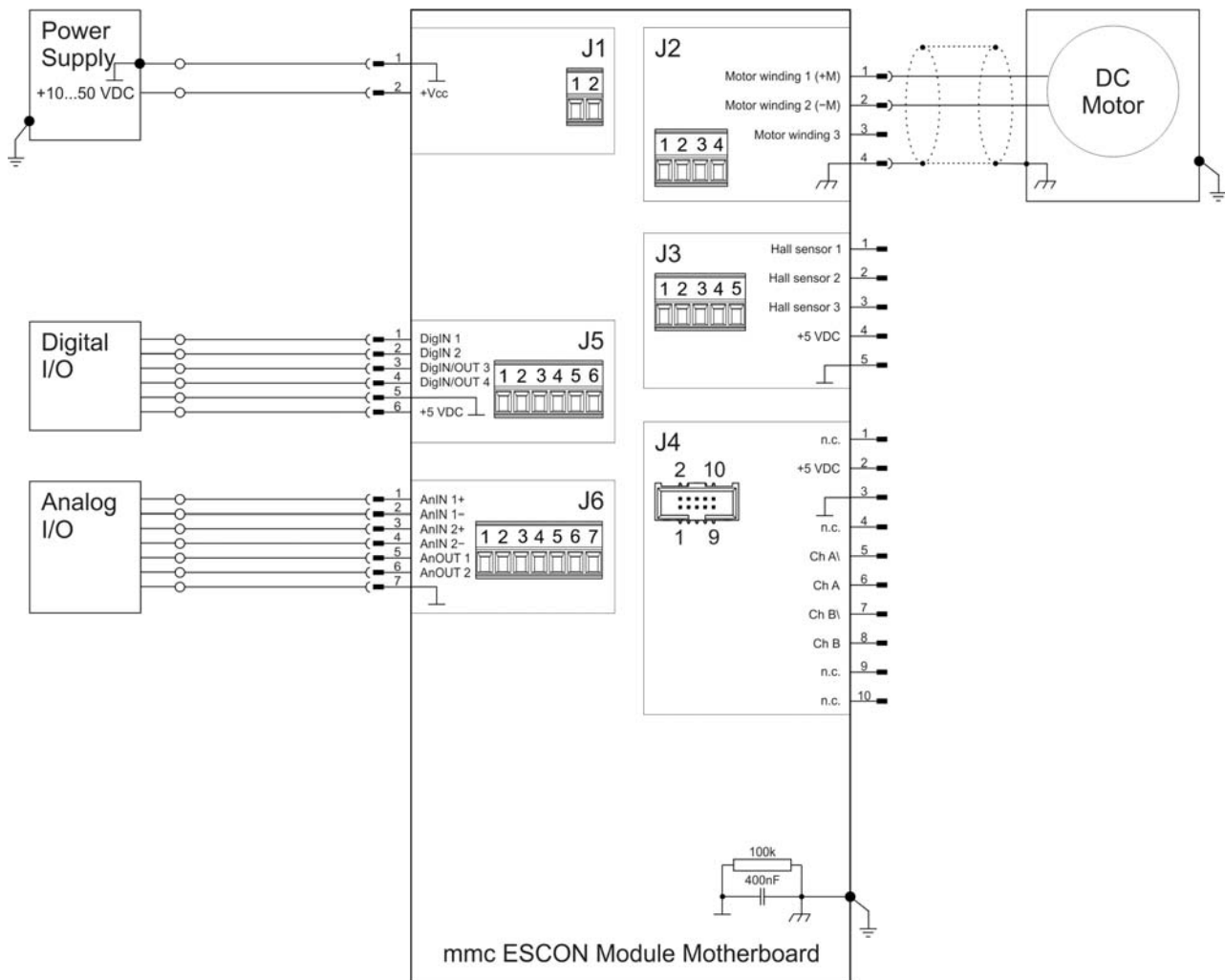


Figure 5-36 ESCON Module MoBo – maxon DC motor (J2)

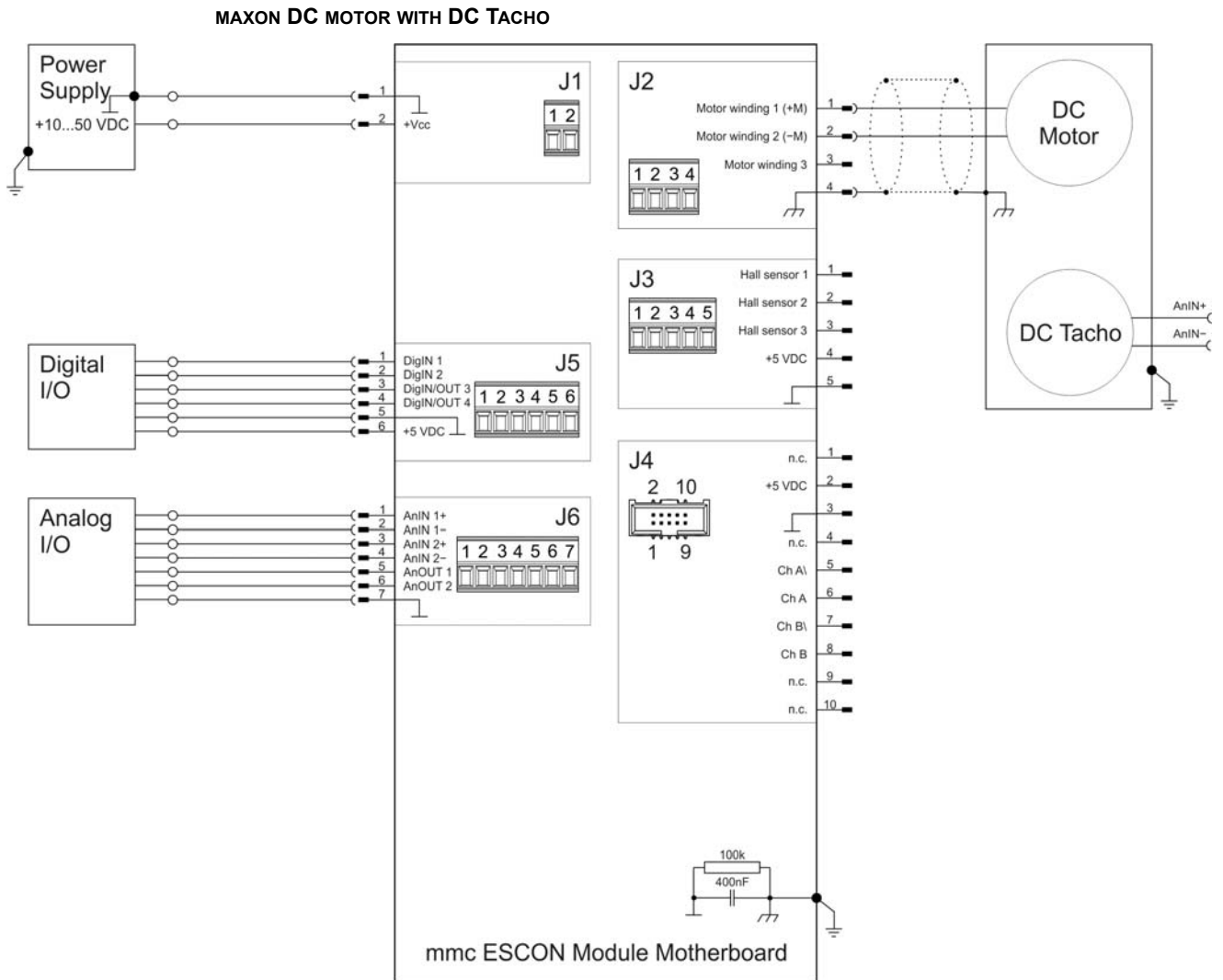


Figure 5-37 ESCON Module MoBo – maxon DC motor with DC Tacho (J2)

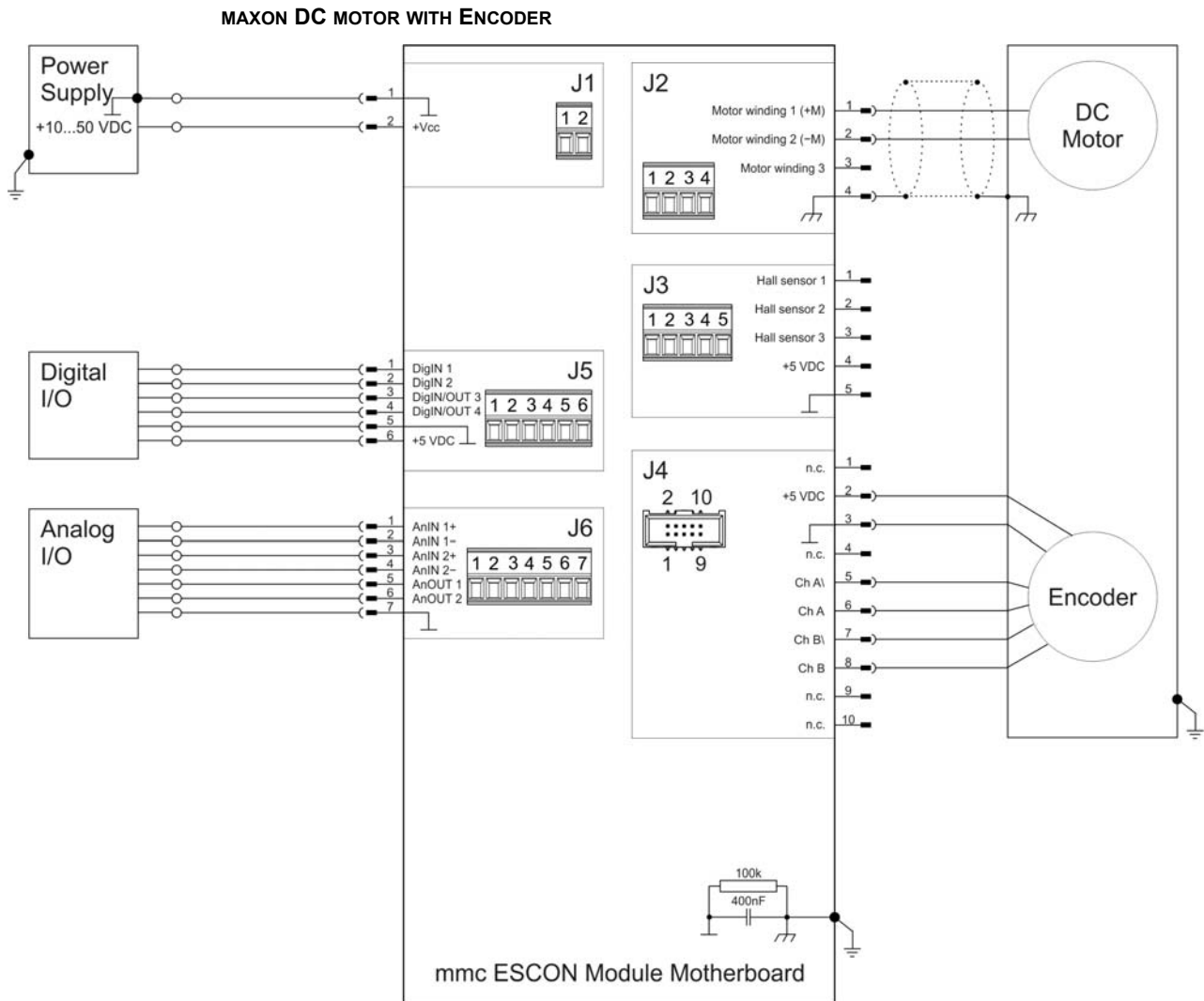


Figure 5-38 ESCON Module MoBo – maxon DC motor with Encoder (J2 / J4)



## MAXON EC MOTOR WITH HALL SENSORS

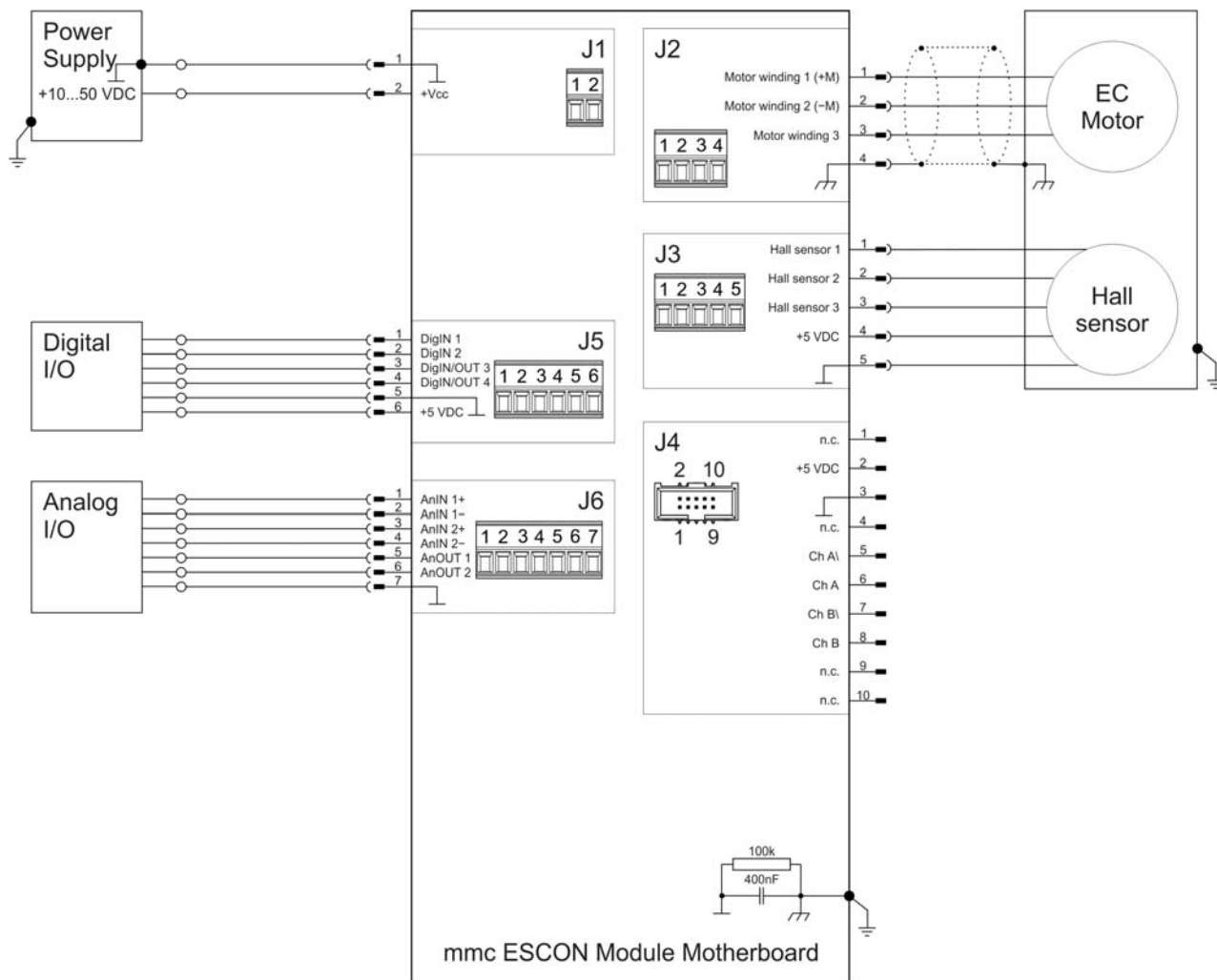


Figure 5-39 ESCON Module MoBo – maxon EC motor with Hall Sensors (J2 / J3)

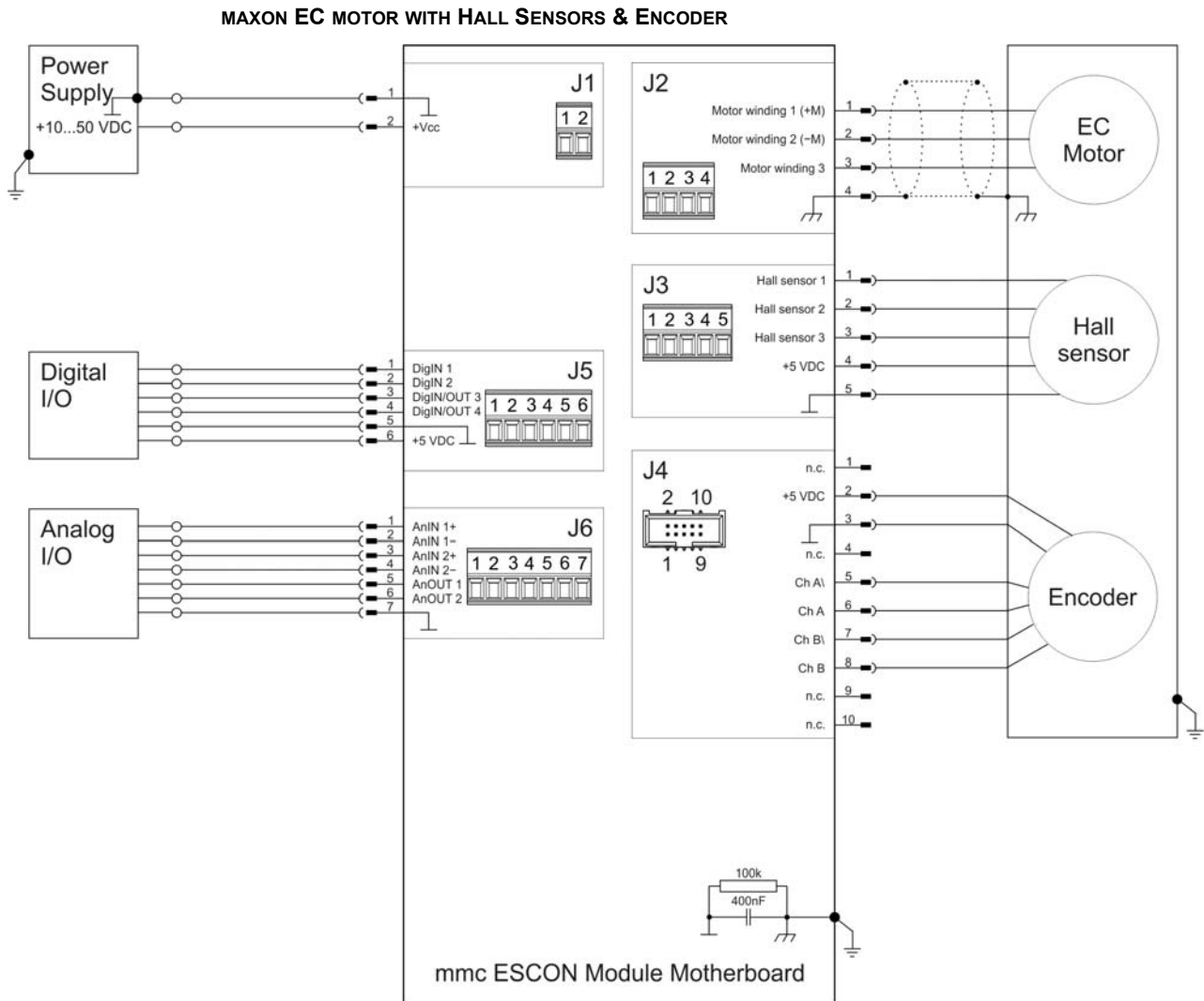


Figure 5-40 ESCON Module MoBo – maxon EC motor with Hall Sensors & Encoders (J2 / J3 / J4)

## 5.8 Spare Parts

| Order number | Description                                                             |
|--------------|-------------------------------------------------------------------------|
| 444144       | 2-pole pluggable screw-type terminal block, 3.5 mm pitch, labeled 1...2 |
| 444145       | 4-pole pluggable screw-type terminal block, 3.5 mm pitch, labeled 1...4 |
| 444146       | 5-pole pluggable screw-type terminal block, 3.5 mm pitch, labeled 1...5 |
| 444147       | 6-pole pluggable screw-type terminal block, 3.5 mm pitch, labeled 1...6 |
| 444148       | 7-pole pluggable screw-type terminal block, 3.5 mm pitch, labeled 1...7 |

Table 5-28 Spare Parts List

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**maxon motor ag**

Brünigstrasse 220  
P.O.Box 263  
CH-6072 Sachseln  
Switzerland

Phone +41 41 666 15 00

Fax +41 41 666 16 50

[www.maxonmotor.com](http://www.maxonmotor.com)